

Continuous-state robust CVaR portfolio optimization via grid-restricted constraint generation

Madani Bezoui · Thiziri Sifaoui · Ahcene Bounceur

Abstract Standard robust portfolio optimization models represent market conditions through discrete regimes or unconditional ambiguity sets. We instead formulate a continuous-state robust portfolio model that minimizes Conditional Value-at-Risk (CVaR) uniformly over a continuously indexed family of kernel-weighted empirical conditional return distributions, with states indexed by the logarithm of the CBOE Volatility Index (log-VIX) and market drawdown. We establish well-posedness, continuity and convexity of the continuous formulation and a Lipschitz bound on the grid-discretization error. For computation we employ a grid-restricted separation oracle and adaptive constraint generation. Across 40 rolling windows, the master problem retains only 3.3–3.4 active state blocks on average as the candidate grid grows from 121 to 961 states, while exchange runtime grows substantially more slowly than the dense extensive-form LP. In a rolling out-of-sample study of US industry portfolios, 1994–2026, the strategy achieves competitive long-run wealth, but a paired circular moving-block percentile bootstrap provides no evidence of a statistically significant Sharpe-ratio advantage over nominal CVaR. The findings highlight the importance of data support in tail regions and the tractability of a grid-restricted approximation to continuous-state robustification.

Keywords Robust Portfolio Optimization · Conditional Value-at-Risk · Semi-Infinite Programming · Kernel-Weighted Conditional Distributions · Constraint Generation

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1 Introduction

The classical mean-variance framework introduced by [21] established the quantitative foundation of modern portfolio theory. However, the practical implementation of mean-variance optimization has long been hindered by its extreme sensitivity to parameter estimation errors. Small perturbations in expected returns or covariance estimates often produce erratic allocations, excessive portfolio turnover, and severe out-of-sample disappointment. To address these vulnerabilities, researchers have developed robust portfolio optimization techniques that optimize allocations against the worst-case parameters within prespecified uncertainty sets [15, 17]. Concurrently, various exact, iterative, and metaheuristic methods have been developed to handle complex real-world constraints in portfolio selection [6, 28], as well as in broader multi-objective manufacturing and supply chain contexts [7].

Traditional robust portfolio models predominantly focus on unconditional uncertainty sets constructed around historical sample moments or empirical distributions [12, 22]. While these models effectively mitigate sensitivity to estimation noise, they typically treat market dynamics as stationary and homogeneous across time. In reality, financial markets exhibit pronounced state-dependent behavior. During tranquil market environments, asset correlations remain moderate and return dispersions offer meaningful diversification benefits. In contrast, during periods of acute financial distress, volatility spikes, asset correlations tend to increase, and tail risks materialize simultaneously across sectors.

To incorporate state dependence, a common practice in financial econometrics is to employ discrete regime-switching models [2]. These models segment market history into a small, finite collection of latent states, such as low-volatility expansionary regimes and high-volatility contractionary regimes. While finite regime classifications offer conceptual simplicity and computational tractability, they impose a discretization on market dynamics. Often, market stress evolves along a continuous spectrum of financial-market conditions. An abrupt jump between two discrete regimes may fail to capture intermediate stress levels and boundary dynamics where vulnerability to systemic shocks varies more continuously.

In this context, our work investigates how a continuous-state robust formulation compares structurally and empirically to standard unconditional and discrete-regime robust portfolio models. We explore whether adaptive constraint generation can effectively scale continuous-state portfolio robustification to overcome the computational limits of dense enumeration. Furthermore, we examine whether state-domain continuous robustification improves out-of-sample risk-adjusted returns and downside protection across non-stationary market environments, and we analyze the role of model regularization—such as bandwidth scaling and empirical support thresholding—in mitigating boundary scenario overfitting within continuous robust sets.

In this paper, we propose a continuous-state robust portfolio optimization framework that bridges non-parametric conditional risk modeling and semi-infinite optimization. Rather than partitioning market history into discrete regimes, we define a continuous, compact state space $\mathcal{U} \subset \mathbb{R}^2$ spanned by observable financial indicators: the logarithm of the CBOE Volatility Index (log-VIX) [9] and the trailing equity market drawdown. Every point $\theta \in \mathcal{U}$ represents a distinct market environment and induces a well-defined conditional empirical return distribution constructed via multivariate Nadaraya–Watson kernel weighting [23, 34].

The investor seeks an allocation that minimizes the worst-case Conditional Value-at-Risk (CVaR) over the entire continuous state space $\mathcal{U}$:

$$\min_{w \in W} \max_{\theta \in \mathcal{U}} \Phi_{\tau}(w, \theta) \quad (1)$$

where W is the capped portfolio simplex and $\Phi_{\tau}(w, \theta)$ is the conditional CVaR at confidence level $1 - \tau = 0.95$ under the empirical measure induced by state θ . We emphasize that this model is a *state-robust* portfolio model, not distributionally robust optimization. For each state θ , the conditional distribution is fixed by the kernel weights, and no distributional ambiguity set is placed around that conditional distribution. Furthermore, the term “adaptive” in our framework refers specifically to *adaptive constraint generation* via a semi-infinite exchange algorithm, rather than a decision policy $w(y_t)$ dynamically conditioned on the current state. The optimal allocation w immunizes against all states in $\mathcal{U}$, while dynamic reallocation over time results from rolling-window re-estimation of the return-state relationship.

Because the state space $\mathcal{U}$ is uncountable, this formulation gives rise to a Semi-Infinite Program (SIP) with an infinite collection of joint risk constraints. Direct solution via dense spatial discretization becomes computationally prohibitive and scales poorly in high dimensions. To solve a finite grid-restricted approximation of the continuous-state SIP efficiently, we use an adaptive exchange algorithm. The method alternates between solving a finite master linear program over a sparse active subset of grid states and executing a full-grid separation oracle that identifies the most adverse state on the candidate grid for the current portfolio.

This paper makes three contributions. First, we introduce a state-robust portfolio formulation in which robustness is imposed over a continuum of covariate-indexed kernel-weighted empirical conditional return distributions, rather than over a distributional ambiguity set or a finite collection of regimes. This construction produces a semi-infinite CVaR optimization problem in which a single allocation is protected against all admissible market states. Second, we establish well-posedness, continuity, convexity and grid-approximation properties of the resulting formulation, and develop a grid-restricted exchange algorithm that exploits the small number of state constraints generated in practice. Third, using a 31.5-year rolling out-of-sample experiment, we show that the method is computationally tractable and economically competitive in wealth terms, while not delivering statistically significant Sharpe-ratio improvements over simpler CVaR benchmarks. The empirical results therefore identify the principal benefit of continuous-state robustness as structural and stabilizing rather than as unconditional performance dominance. We do not claim novelty for exchange algorithms as such; the contribution lies in the continuous-state construction, its regularity analysis, and the demonstration that the induced semi-infinite problem is sparse in practice.

To facilitate open science, reproducibility and independent verification of all empirical findings, the complete Julia and Python codebase, the cleaned financial datasets, and all figure generation routines are maintained in an open-source repository [8].

The remainder of this manuscript is structured as follows. Section 2 reviews related literature across robust optimization, conditional stochastic programming, and semi-infinite algorithms. Sections 3 and 4 develop the continuous-state modeling framework, kernel weighting mechanics, and theoretical existence theorems. Section 5 presents the adaptive exchange algorithm and distinguishes exact continuous

separation from the grid-restricted separation used in the numerical implementation. Section 6 details the rolling-window backtesting protocol, while Sections 7 and 8 report empirical and computational results across approximately 31.5 years. Section 9 presents model-specification sensitivity analysis, and Section 10 provides discussion, practical limitations, and concluding remarks.

2 Related work

Our methodology intersects three active streams of quantitative operations research: robust portfolio selection, conditional stochastic programming with side information, and semi-infinite optimization algorithms.

2.1 Classical and distributionally robust portfolio selection

Classical robust portfolio selection addresses the sensitivity of Markowitz mean-variance optimization to estimation noise by assuming that expected return vectors and covariance matrices reside in deterministic uncertainty sets [15, 17]. Concurrently, various exact, iterative, and metaheuristic methods have been developed to handle complex real-world constraints in portfolio selection [6, 28], as well as in broader multi-objective manufacturing and supply chain contexts [7]. These formulations yield second-order cone or semidefinite programs that provide worst-case guarantees. However, moment-based robust models frequently rely on implicit symmetry assumptions that fail to reflect the severe skewness, heavy tails, and asymmetric dependence structures inherent in financial return distributions.

To capture tail risk directly, [29, 30] introduced the Conditional Value-at-Risk (CVaR) as a coherent risk measure that can be optimized efficiently via linear programming. Building on this foundation, distributionally robust optimization (DRO) seeks to minimize risk over an ambiguity set of probability distributions constructed around an empirical baseline. Modern DRO frameworks define ambiguity balls using moment constraints [12] or optimal transport metrics such as the Wasserstein distance [22].

Recent theoretical developments in DRO have expanded its algorithmic scope. For instance, [11] propose tractable regularized reformulations of Wasserstein DRO problems that achieve global convergence, while [1] analyzes two-stage distributionally robust programs with finite supports. In related structural contexts, [35] draw geometric connections between ambiguity sets and covariance shrinkage, and [33] study piecewise affine decision rules for robust multi-stage settings.

2.2 Conditional stochastic optimization and covariate side information

A rapidly expanding body of literature investigates how observable side information (covariates) can be integrated into data-driven decision-making. [4] established a general framework for predictive prescriptions, demonstrating that non-parametric kernel regression, k -nearest neighbors, and tree-based weights can effectively condition decision problems on prevailing covariates. In the context of risk management, [31] examined non-parametric kernel estimation of conditional expected shortfall.

In recent contributions, [27] introduce an integrated conditional estimation-optimization methodology that directly couples regression trees with downstream optimization tasks, minimizing decision error rather than prediction loss. [3] explore phase transitions in data-driven decision-making, providing optimal sample complexity bounds for prescriptive models with side information. Earlier foundational frameworks in conditional DRO include [14], who developed distributionally robust stochastic programs with side information based on trimmed Wasserstein balls, and [24], who formulated robust conditional portfolio decisions via optimal transport metrics centered on covariate-conditioned empirical distributions.

Closest in spirit to the present computational setting, [10] learn contextual uncertainty sets and solve the resulting conditional robust programs by constraint generation; their sets are learned around the observed context, whereas the family indexed here is fixed by the kernel construction and the robustness is taken over the whole state domain rather than around one observed state.

Our framework differs in its mathematical construction from conditional DRO. In conditional DRO, the decision-maker constructs an ambiguity set around the conditional distribution corresponding to the currently observed state y_t . In our continuous-state robust model, we do not restrict attention to a single observed state. Instead, we seek an allocation that remains robust across the entire family of kernel-weighted conditional distributions induced by all states $\theta \in \mathcal{U}$. We refer to this approach as continuous-state robust optimization over kernel-weighted empirical conditional distributions.

To distinguish our approach from standard Conditional Distributionally Robust Optimization (DRO), consider the formal structures:

$$\text{Conditional DRO: } \min_{w \in W} \sup_{P \in \mathcal{B}(\hat{P}_{y_t})} \text{CVaR}_P(w) \quad (2)$$

$$\text{This paper (Robust SIP): } \min_{w \in W} \sup_{\theta \in \mathcal{U}} \text{CVaR}_{\hat{P}_\theta}(w) \quad (3)$$

The robustification operator differs significantly: standard DRO considers a measure-ball $\mathcal{B}$ around a fixed covariate y_t , whereas we take a supremum over a covariate-indexed family of fixed empirical measures over a continuous state space $\mathcal{U}$. Consequently, the optimization structure differs (single-state DRO reformulations versus SIP with a continuum of LP blocks). These approaches are complementary, and one could in principle compose them in future work.

2.3 Semi-infinite programming and exchange algorithms

Semi-infinite programming (SIP) considers optimization problems with a finite number of decision variables and an infinite index set of constraints [18, 20]. SIP techniques have found applications in engineering design, robust control, and spectral estimation, but their operational use in financial portfolio allocation has been constrained by computational complexity.

The standard solution paradigm for convex SIPs is the cutting-plane or exchange algorithm. At each iteration, the master problem solves a relaxed subproblem over a finite subset of constraints, while a separation oracle identifies the most violated constraint across the continuous index set. When the oracle can be solved to

certified global optimality, the exchange algorithm generates monotonically non-decreasing lower bounds and certified upper bounds that converge to the true global optimum. In practical settings where the oracle is solved over a grid or via local search heuristics, exact continuous separation is relaxed. For broader methodological context, [25] formalize the convergence and error propagation of convex SIP algorithms equipped with inexact separation oracles.

Table 1 positions the present framework relative to these strands of the literature, contrasting for each model family the information used to index the market state, the way the conditional distribution is represented, whether the resulting decision is state-dependent, the dimension along which robustness is imposed, and the computational form that must ultimately be solved. In the table, weighted SAA denotes weighted sample average approximation.

Table 1 Comparison of conditional and robust risk models

Paradigm	Objective	Distributional Assumption	Covariate Role
Unconditional Markowitz	Mean-Variance	Stationary Gaussian	None
Nominal	Expected	True conditional	Directly defines the distribution
Conditional	Shortfall	$P(Y X)$ known	
Standard DRO	Worst-case Risk over $\mathcal{P}$	Global ambiguity set $\mathcal{P}$	Typically unconditional
Covariate-driven DRO	Conditional Worst-case Risk	State-dependent ambiguity set $\mathcal{P}(x)$	Centers the conditional ambiguity set
Robust SIP (Proposed)	Worst-case Risk over $\mathcal{U}$	Continuous kernel-weighted empirical	Defines a continuous geometric state space $\mathcal{U}$

3 Model

We consider an investment universe consisting of N risky financial assets over a discrete time horizon $t = 1, \dots, T$. Let $x_t = (x_{1,t}, \dots, x_{N,t})^\top \in \mathbb{R}^N$ denote the vector of realized asset returns from trading day $t - 1$ to day t . The joint conditional distribution of asset returns is modulated by prevailing financial-market state variables observable at time $t - 1$, denoted by the vector $y_{t-1} \in \mathbb{R}^M$.

3.1 State variable selection and compact state space

In this study, we focus on two primary financial indicators that capture distinct dimensions of market risk ($M = 2$): the logarithm of the CBOE Volatility Index, $\log V_t$ (log-VIX), and the Trailing Market Drawdown (D_t) representing the peak-to-trough decline of the broad equity market index over a trailing quarterly lookback window of $L = 63$ trading days:

$$D_t = 1 - \frac{P_t}{\max_{s \in [t-L, t]} P_s} \in [0, 1] \quad (4)$$

where P_t is the cumulative total return index of the broad market proxy (computed as the equal-weighted composite of the 30 industry portfolios). The choice of an equal-weighted proxy rather than a value-weighted aggregate is deliberate: the equal-weighted index is more sensitive to distress in smaller sectors and therefore serves as a more responsive indicator of broad-based distress across the entire opportunity set. By construction, $D_t \geq 0$ measures positive drawdown severity (0% representing peak wealth, and positive values indicating drawdown depth).

The state vector at time $t-1$ is $y_{t-1} = (\log V_{t-1}, D_{t-1})^\top \in \mathbb{R}^2$. For each rolling training window, define

$$v_{\min} = \min_t \log V_{t-1}, \quad v_{\max} = \max_t \log V_{t-1}, \quad (5)$$

and

$$d_{\min} = \min_t D_{t-1}, \quad d_{\max} = \max_t D_{t-1}. \quad (6)$$

All extrema, variances, bandwidths, and grid boundaries are recomputed exclusively from the current 1,260-day training window. We define the continuous state space $\mathcal{U} \subset \mathbb{R}^2$ as a compact rectangle containing all historically observed market states, augmented by a fixed 10% outer margin to examine states slightly beyond the observed training-window extrema, while strictly respecting the non-negative domain of drawdown:

$$\mathcal{U} = [v_{\min} - \delta_v, v_{\max} + \delta_v] \times [\max\{0, d_{\min} - \delta_d\}, \min\{1, d_{\max} + \delta_d\}] \quad (7)$$

where $\delta_v = 0.10(v_{\max} - v_{\min})$ and $\delta_d = 0.10(d_{\max} - d_{\min})$. The lower bound on drawdown is clamped at 0, preventing negative drawdown values. By construction, $\mathcal{U}$ is a closed and bounded subset of $\mathbb{R}^2$, ensuring compactness.

Remark 1 (Domain rectangularity and empirical support) The Cartesian bounding box $\mathcal{U}$ treats every combination of VIX and drawdown as plausible. However, certain corners (e.g., very high VIX with zero drawdown) have low empirical support. In our empirical analysis, we track the Effective Sample Size of active states to monitor local data support, and we evaluate regularized uncertainty sets $\mathcal{U}_{\text{supp}} = \{\theta \in \mathcal{U} : \text{ESS}(\theta) \geq E_{\min}\}$ in Section 9.

3.2 Kernel-weighted conditional distributions and effective sample size

For any continuous state query $\theta = (v_\theta, d_\theta)^\top \in \mathcal{U}$, we construct a state-dependent empirical probability distribution over historical return realizations $\{x_1, \dots, x_T\}$. Each historical return observation x_t is assigned a Nadaraya–Watson kernel weight [23, 34] based on the proximity of its preceding state y_{t-1} to the target state θ :

$$p_t(\theta) = \frac{K_H(y_{t-1} - \theta)}{\sum_{s=1}^T K_H(y_{s-1} - \theta)} = \frac{\exp\left(-\frac{1}{2}(y_{t-1} - \theta)^\top H^{-1}(y_{t-1} - \theta)\right)}{\sum_{s=1}^T \exp\left(-\frac{1}{2}(y_{s-1} - \theta)^\top H^{-1}(y_{s-1} - \theta)\right)} \quad (8)$$

where $H \in \mathbb{R}^{2 \times 2}$ is a symmetric positive-definite bandwidth matrix. We specify H using the bivariate rule-of-thumb [32], defined by marginal bandwidths $h_m = T^{-1/6} \hat{\sigma}_m$:

$$H = \text{diag}\left(h_{\log\text{-VIX}}^2, h_{\text{DD}}^2\right) = T^{-1/3} \text{diag}\left(\hat{\sigma}_{\log\text{-VIX}}^2, \hat{\sigma}_{\text{DD}}^2\right) \quad (9)$$

for $M = 2$, where $\hat{\sigma}_{\log\text{-VIX}}^2$ and $\hat{\sigma}_{\text{DD}}^2$ are the sample variances of log-VIX and drawdown, respectively, computed over the current training window. The diagonal bandwidth matrix scales the two state dimensions by their respective sample variances but does not account for their empirical cross-covariance. While log-VIX and drawdown exhibit non-trivial empirical correlation, using a diagonal matrix acts as a form of geometric regularization that avoids excessively narrow, elongated kernel contours that could overfit the historical dependence structure. Future work could explore full-covariance bandwidth selection methods to further refine the shape of the conditional empirical measure.

Because the Gaussian kernel is strictly positive everywhere on $\mathbb{R}^2$, the normalization denominator is strictly positive for all $\theta \in \mathcal{U}$. Consequently, the kernel weights satisfy $p_t(\theta) > 0$ and $\sum_{t=1}^T p_t(\theta) = 1$ for every $\theta \in \mathcal{U}$. Each continuous state θ induces a well-defined conditional empirical distribution $\mathbb{P}(X|\theta) = \sum_{t=1}^T p_t(\theta)\delta_{x_t}$ (as illustrated in Figure 1), where δ_{x_t} denotes the Dirac delta mass at historical return vector x_t .

To monitor local data support across the continuous domain, we compute the Effective Sample Size (ESS) for any state $\theta \in \mathcal{U}$:

$$\text{ESS}(\theta) = \frac{1}{\sum_{t=1}^T p_t(\theta)^2} \quad (10)$$

When weights are uniformly distributed ($p_t(\theta) = 1/T$), the ESS reaches its theoretical maximum $\text{ESS}(\theta) = T$. Conversely, if all weight concentrates on a single historical observation, $\text{ESS}(\theta) = 1$. For descriptive purposes, we report $\tau \times \text{ESS}(\theta)$ as a coarse tail-support proxy. Because kernel weights are non-uniform and tail membership depends on the ordering of portfolio losses, this quantity should not be interpreted as a formal tail effective sample size.

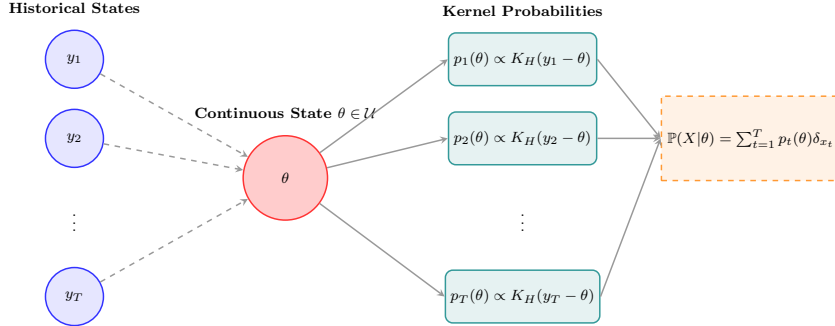


Fig. 1 Continuous market state mapping using kernel weighting

We emphasize that mathematical well-posedness and statistical reliability are distinct properties. The strict positivity of the Gaussian kernel ensures that $\hat{P}_\theta$ is defined for every $\theta \in \mathcal{U}$, but it does not imply that every conditional distribution is estimated with comparable precision. In low-support regions the conditional empirical measure may effectively rest on a small number of historical observations, so a state can be perfectly well defined and still be poorly estimated. We therefore use the effective sample size as a diagnostic of local estimation reliability, and study ESS-restricted state domains explicitly in Section 9.

4 Theory

Let $w = (w_1, \dots, w_N)^\top \in \mathbb{R}^N$ denote the portfolio weight vector. The portfolio loss under realized return vector x_t is $-x_t^\top w$. We define the feasible portfolio set $W \subset \mathbb{R}^N$ under standard institutional long-only constraints with an individual weight cap $\bar{w} = 0.15$ and a target expected return constraint:

$$W = \left\{ w \in \mathbb{R}^N : \sum_{i=1}^N w_i = 1, \quad 0 \leq w_i \leq \bar{w} \quad \forall i = 1, \dots, N, \quad \hat{\mu}^\top w \geq \mu_{\text{target}} \right\} \quad (11)$$

where $\hat{\mu} = \frac{1}{T} \sum_{t=1}^T x_t$ is the unconditional sample mean return vector, μ_{target} is the required expected return, and $\bar{w} = 0.15$ is the maximum allocation to any single asset.

Proposition 1 (Feasibility of the capped portfolio set) *Let $S_{\bar{w}} = \{w \in \mathbb{R}^N : \mathbf{1}^\top w = 1, 0 \leq w_i \leq \bar{w}\}$. Then $S_{\bar{w}} \neq \emptyset$ if and only if $N\bar{w} \geq 1$. Conditional on this, let $\hat{\mu}_{(1)} \geq \dots \geq \hat{\mu}_{(N)}$ be the components of $\hat{\mu}$ sorted in descending order, and $\hat{\mu}_{[1]} \leq \dots \leq \hat{\mu}_{[N]}$ be sorted in ascending order. Define $q = \lfloor 1/\bar{w} \rfloor$ and the residual weight $r = 1 - q\bar{w}$. The maximum and minimum expected returns attainable in $S_{\bar{w}}$ are respectively $\mu_{\max} = \bar{w} \sum_{i=1}^q \hat{\mu}_{(i)} + r\hat{\mu}_{(q+1)} \mathbb{1}_{\{r>0\}}$ and $\mu_{\min} = \bar{w} \sum_{i=1}^q \hat{\mu}_{[i]} + r\hat{\mu}_{[q+1]} \mathbb{1}_{\{r>0\}}$, with the residual term omitted when $r = 0$. The attainable expected-return set is $[\mu_{\min}, \mu_{\max}]$. Then $W \neq \emptyset$ if and only if $\mu_{\text{target}} \leq \mu_{\max}$. Moreover, there exists $w \in S_{\bar{w}}$ satisfying $\hat{\mu}^\top w > \mu_{\text{target}}$ if and only if $\mu_{\text{target}} < \mu_{\max}$.*

Proof If $N\bar{w} < 1$, then

$$\sum_{i=1}^N w_i \leq N\bar{w} < 1, \quad (12)$$

so the capped simplex is empty. If $N\bar{w} \geq 1$, a feasible fully invested allocation exists. The continuous knapsack problem

$$\max\{\hat{\mu}^\top w : \mathbf{1}^\top w = 1, 0 \leq w_i \leq \bar{w}\} \quad (13)$$

is solved by allocating $\bar{w}$ to the assets with the largest sample means and assigning the residual weight r to the next asset, which yields $\mu_{\max}$. Analogously, allocating weight first to the assets with the smallest sample means yields $\mu_{\min}$. Since the capped simplex is convex and $w \mapsto \hat{\mu}^\top w$ is continuous and linear, its image is the interval $[\mu_{\min}, \mu_{\max}]$. Therefore, $W \neq \emptyset$ if and only if $\mu_{\text{target}} \leq \mu_{\max}$, and strict feasibility of the return constraint holds if and only if $\mu_{\text{target}} < \mu_{\max}$. Finally, W is a closed convex subset of the compact capped simplex and is therefore compact and convex.

Following [29], the conditional CVaR evaluated at state θ at confidence level $\alpha = 1 - \tau$ (with $\tau = 0.05$) is:

$$\Phi_\tau(w, \theta) = \min_{z \in \mathbb{R}} \left\{ z + \frac{1}{\tau} \sum_{t=1}^T p_t(\theta) [-x_t^\top w - z]_+ \right\} \quad (14)$$

where $z \in \mathbb{R}$ represents the conditional Value-at-Risk (VaR) at level $\alpha = 1 - \tau$, and $[u]_+ = \max(0, u)$. The optimal VaR minimizer satisfies $z^*(\theta) \in \text{VaR}_{1-\tau}(\ell(w) \mid \theta)$.

Although $z^*(\theta)$ may be a non-singleton interval for discrete empirical distributions, the minimum value $\Phi_\tau(w, \theta)$ is uniquely defined.

The continuous-state robust portfolio optimization problem seeks an allocation $w \in W$ that minimizes the worst-case conditional CVaR across all market states in $\mathcal{U}$:

$$v^* = \min_{w \in W} \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta) \quad (15)$$

Introducing an auxiliary epigraph scalar $\eta \in \mathbb{R}$ representing the robust risk level, the problem is formulated as the following convex Semi-Infinite Program (SIP):

$$\min_{w \in \mathbb{R}^N, \eta \in \mathbb{R}} \quad \eta \quad (16)$$

$$\text{s.t.} \quad \Phi_\tau(w, \theta) \leq \eta \quad \forall \theta \in \mathcal{U} \quad (17)$$

$$\sum_{i=1}^N w_i = 1 \quad (18)$$

$$0 \leq w_i \leq \bar{w} \quad \forall i = 1, \dots, N \quad (19)$$

$$\hat{\mu}^\top w \geq \mu_{\text{target}} \quad (20)$$

Proposition 2 (Compactness, continuity, and convexity) *Assume that $H \succ 0$, $N\bar{w} \geq 1$, and $\mu_{\text{target}} \leq \mu_{\text{max}}$. Then:*

1. *The feasible portfolio set W and the market state space $\mathcal{U}$ are non-empty, compact subsets of $\mathbb{R}^N$ and $\mathbb{R}^2$, respectively.*
2. *For every $t = 1, \dots, T$, the kernel weight function $\theta \mapsto p_t(\theta)$ is infinitely differentiable and strictly positive on $\mathcal{U}$.*
3. *The conditional CVaR function $(w, \theta) \mapsto \Phi_\tau(w, \theta)$ is jointly continuous on $W \times \mathcal{U}$.*
4. *For every fixed $\theta \in \mathcal{U}$, the function $w \mapsto \Phi_\tau(w, \theta)$ is real-valued and convex on W .*

Proof Item 1 follows from Proposition 1 and the closed bounding box construction of $\mathcal{U}$. For Item 2, the Gaussian kernel $K_H(u)$ is smooth and strictly positive on $\mathbb{R}^2$. The normalization denominator $\sum_{s=1}^T K_H(y_{s-1} - \theta)$ is strictly positive on compact $\mathcal{U}$. Thus $p_t(\theta)$ is a ratio of smooth, positive functions with a non-vanishing denominator, making it smooth and strictly positive.

For Item 3, define

$$F(w, z, \theta) = z + \frac{1}{\tau} \sum_{t=1}^T p_t(\theta) [-x_t^\top w - z]_+. \quad (21)$$

Since $p_t(\theta)$ and $(w, z) \mapsto [-x_t^\top w - z]_+$ are continuous, F is jointly continuous on $\mathbb{R}^N \times \mathbb{R} \times \mathcal{U}$. For every $w \in W$, a weighted empirical VaR minimizer can be selected in

$$\left[\min_t \{-x_t^\top w\}, \max_t \{-x_t^\top w\} \right]. \quad (22)$$

Since W is compact and the return sample is finite, all such minimizers lie in the common compact interval $[-M_L, M_L]$, where

$$M_L = \max_{w \in W, t} |x_t^\top w| < \infty. \quad (23)$$

By Berge's Maximum Theorem, the partial minimum $\Phi_\tau(w, \theta) = \min_{z \in [-M_L, M_L]} F(w, z, \theta)$ is continuous on $W \times \mathcal{U}$.

For Item 4, since $F(w, z, \theta)$ is jointly convex in (w, z) for each fixed θ , the partial minimization over convex z preserves convexity in w . Hence, $w \mapsto \Phi_\tau(w, \theta)$ is convex on W for each $\theta \in \mathcal{U}$.

Theorem 1 (Existence of optimal robust allocation) *Under the assumptions of Proposition 2, the robust objective function $G(w) = \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta)$ is continuous and convex on W . Furthermore, the supremum is attained for every $w \in W$, and the continuous-state robust portfolio optimization problem admits an optimal solution $w^* \in W$.*

Proof Because $\mathcal{U}$ is compact and $\Phi_\tau(w, \cdot)$ is continuous on $\mathcal{U}$ for every $w \in W$, the Extreme Value Theorem guarantees that the supremum is achieved at some $\theta^*(w) \in \mathcal{U}$, so $G(w) = \max_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta)$. As the pointwise maximum of a family of convex functions $\{w \mapsto \Phi_\tau(w, \theta)\}_{\theta \in \mathcal{U}}$, $G(w)$ is convex on W . Joint continuity of $\Phi_\tau(w, \theta)$ on compact $W \times \mathcal{U}$ implies by Berge's Maximum Theorem that $G(w)$ is continuous on W . Finally, minimizing the continuous function $G(w)$ over the non-empty compact set W guarantees the existence of an optimal solution $w^* \in W$ by the Weierstrass Theorem.

5 Algorithms

A full Cartesian-grid approximation of the continuous state space $\mathcal{U}$ yields a large-scale linear program with one CVaR state block for every candidate grid point. We design an adaptive exchange algorithm that iteratively generates only the binding stress states.

5.1 Master problem and separation oracle

At iteration $k \geq 1$, the master problem optimizes portfolio weights w and the robust risk level η subject to a finite active subset of grid states $\hat{\mathcal{U}}_k = \{\theta^{(1)}, \dots, \theta^{(m_k)}\} \subseteq \hat{\mathcal{U}} \subset \mathcal{U}$. For each active state $\theta^{(j)} \in \hat{\mathcal{U}}_k$, we introduce a state-specific VaR variable $z_{\theta^{(j)}} \in \mathbb{R}$ and scenario excess loss variables $u_{t, \theta^{(j)}} \geq 0$. The master problem is the following finite Linear Program (LP):

$$\text{LB}_k = \min_{w, \eta, \{z_\theta\}, \{u_{t, \theta}\}} \eta \quad (24)$$

$$\text{s.t.} \quad z_\theta + \frac{1}{\tau} \sum_{t=1}^T p_t(\theta) u_{t, \theta} \leq \eta \quad \forall \theta \in \hat{\mathcal{U}}_k \quad (25)$$

$$u_{t, \theta} \geq -x_t^\top w - z_\theta \quad \forall t = 1, \dots, T, \forall \theta \in \hat{\mathcal{U}}_k \quad (26)$$

$$u_{t, \theta} \geq 0 \quad \forall t = 1, \dots, T, \forall \theta \in \hat{\mathcal{U}}_k \quad (27)$$

$$\sum_{i=1}^N w_i = 1, \quad 0 \leq w_i \leq \bar{w} \quad \forall i = 1, \dots, N \quad (28)$$

$$\hat{\mu}^\top w \geq \mu_{\text{target}} \quad (29)$$

Solving the master LP yields a candidate allocation w^k and a lower bound LB_k on both the grid-restricted optimum $\hat{v}^*$ and the continuous optimum v^* , because $\hat{\mathcal{U}}_k \subseteq \hat{\mathcal{U}} \subset \mathcal{U}$.

Given w_k , the separation oracle searches for the most adverse state that maximizes conditional CVaR. In our empirical implementation, the separation step searches a candidate discretization $\hat{\mathcal{U}} \subset \mathcal{U}$ ($21 \times 21 = 441$ states):

$$\theta^* = \arg \max_{\theta \in \hat{\mathcal{U}}} \Phi_\tau(w^k, \theta), \quad \hat{G}_k = \Phi_\tau(w^k, \theta^*) \quad (30)$$

For computational tractability, we replace the continuous domain $\mathcal{U}$ with a finite candidate grid $\hat{\mathcal{U}}$, resulting in the grid-restricted robust approximation:

$$\hat{v}^* = \min_{w \in W} \max_{\theta \in \hat{\mathcal{U}}} \Phi_\tau(w, \theta) \quad (31)$$

Every numerical “optimal”, “worst-case”, or “binding” claim regarding our adaptive exchange algorithm refers to $\hat{v}^*$ over $\hat{\mathcal{U}}$, not the continuous domain $\mathcal{U}$.

Because the portfolio allocation w^k is fixed during separation, the conditional CVaR values $\Phi_\tau(w^k, \theta)$ for all $\theta \in \hat{\mathcal{U}}$ are evaluated in overall $O(T \log T + KT)$ time per oracle call: sorting the T portfolio losses costs $O(T \log T)$ once per iteration, and for each of the $K = |\hat{\mathcal{U}}|$ grid states, evaluating the kernel weights and accumulating the weighted tail expectation via a running sum over the sorted losses costs $O(T)$, hence $O(KT)$ across the grid.

$$\ell_t = -x_t^\top w^k, \quad \ell_{(1)} \leq \dots \leq \ell_{(T)}, \quad (32)$$

and reordering the kernel probabilities accordingly as $p_{(t)}(\theta)$. The overall adaptive exchange algorithm is depicted in Figure 2.

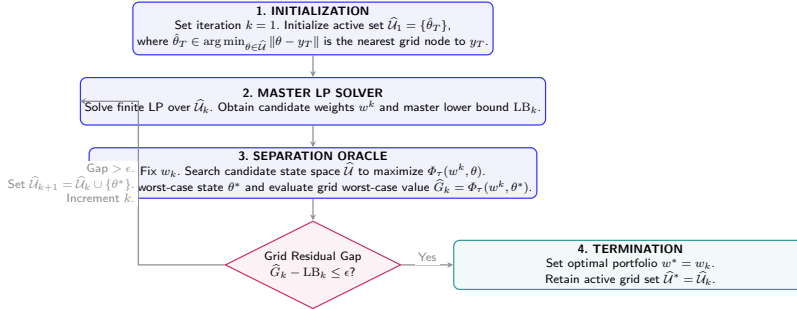


Fig. 2 Grid-restricted adaptive constraint-generation algorithm

5.2 Exact continuous separation and grid-restricted separation

We first recall the convergence guarantee obtained under exact continuous separation and then state the scope of the finite grid-restricted implementation.

Proposition 3 (Convergence under exact separation) *Assume that each master problem is solved to exact LP optimality and the separation oracle identifies the worst-case state $\theta^* = \arg \max_{\theta \in \mathcal{U}} \Phi_\tau(w^k, \theta)$ with exact value $G(w^k) = \Phi_\tau(w^k, \theta^*)$. Then:*

1. The master lower bounds form a non-decreasing sequence: $\text{LB}_1 \leq \text{LB}_2 \leq \dots \leq v^*$.
2. The oracle values satisfy $G(w_k) \geq v^*$ for all $k \geq 1$.
3. For any convergence tolerance $\epsilon > 0$, the exchange algorithm terminates in a finite number of iterations with an ϵ -optimal portfolio $w^k \in W$ satisfying:

$$0 \leq G(w^k) - v^* \leq \epsilon, \quad \sup_{\theta \in \mathcal{U}} [\Phi_\tau(w^k, \theta) - \text{LB}_k] \leq \epsilon \quad (33)$$

Proof Because $\hat{\mathcal{U}}_k \subseteq \hat{\mathcal{U}}_{k+1} \subseteq \mathcal{U}$, the feasible region of the master LP shrinks monotonically, implying $\text{LB}_k \leq \text{LB}_{k+1} \leq v^*$. For Item 2, $G(w^k) = \sup_{\theta \in \mathcal{U}} \Phi_\tau(w^k, \theta) \geq \min_{w \in W} \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta) = v^*$.

For Item 3, upon termination with $G(w^k) - \text{LB}_k \leq \epsilon$, we have $0 \leq G(w^k) - v^* \leq G(w^k) - \text{LB}_k \leq \epsilon$, and the portfolio w^k satisfies all finite constraints in W while the epigraph pair $(w^k, \text{LB}_k + \epsilon)$ is feasible for the continuous constraint set.

To establish finite termination, suppose for contradiction that the algorithm does not terminate. Then it generates an infinite sequence of states $\{\theta_k\}_{k=1}^\infty \subset \mathcal{U}$ such that $\Phi_\tau(w^k, \theta_k) - \text{LB}_k > \epsilon$ for all k . For any $j < k$, the constraint for θ_j was included in master problem k , so $\Phi_\tau(w^k, \theta_j) \leq \text{LB}_k$. Thus $\Phi_\tau(w^k, \theta_k) - \Phi_\tau(w^k, \theta_j) > \epsilon$. Since Φ_τ is continuous on the compact set $W \times \mathcal{U}$, it is uniformly continuous. Hence, for every $\epsilon > 0$, there exists $\delta_\epsilon > 0$ such that $\|\theta - \theta'\| < \delta_\epsilon \implies |\Phi_\tau(w, \theta) - \Phi_\tau(w, \theta')| < \epsilon$ uniformly for all $w \in W$. Therefore, $\|\theta_k - \theta_j\| \geq \delta_\epsilon$ for all $j < k$. A compact metric space is totally bounded and therefore cannot contain an infinite δ_ϵ -separated sequence, which contradicts the compactness of $\mathcal{U}$ [18, 20]. This exact-oracle convergence is a classical consequence of standard SIP exchange methods.

Remark 2 (Monotonicity of bounds) While the master lower bounds LB_k are guaranteed to be monotonically non-decreasing, the separation oracle worst-case values $\hat{G}(w^k)$ are not guaranteed to decrease monotonically at every iteration because the portfolio w^k changes. However, the best incumbent worst-case value $\bar{G}_k = \min_{1 \leq j \leq k} \hat{G}(w^j)$ is monotonically non-increasing by construction.

Remark 3 (Grid-restricted separation) When the separation oracle is evaluated over a finite candidate grid $\hat{\mathcal{U}} \subset \mathcal{U}$, it computes

$$\hat{G}(w) = \max_{\theta \in \hat{\mathcal{U}}} \Phi_\tau(w, \theta) \leq \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta). \quad (34)$$

Since Φ_τ is continuous on the compact set $W \times \mathcal{U}$, it is uniformly continuous. Therefore, the discrepancy between the continuous and grid-restricted objectives tends to zero as the grid dispersion radius tends to zero. Consequently, all claims of ‘optimal’ weights and ‘worst-case’ values in the empirical sections refer strictly to grid-restricted optimality over $\hat{\mathcal{U}}$, unless further assessed through the heuristic continuous-domain diagnostics reported in Section 5.4. To ensure the algorithm operates strictly over $\hat{\mathcal{U}}$, the active subset $\hat{\mathcal{U}}_1$ is initialized with the nearest grid node to the most recent observation y_T . Section 5.4 derives a computable bound on L_Φ and reports per-window heuristic observed gaps, and the grid-refinement sensitivity analysis reported in Section 9 provides a complementary assessment. Regarding initialization, re-running the 20-window sensitivity subset with three active-set initializations (nearest node to y_T , the grid center, a five-node spread around the unconditional mean state) yields identical terminal objectives (equal to

within solver tolerance 10^{-8}) and identical iteration counts in 20 of 20 windows, indicating robustness of the exchange path to the initial active set.

The grid-resolution statistic $\rho = \sup_{\theta \in \mathcal{U}} \min_{\hat{\theta} \in \hat{\mathcal{U}}} \|\theta - \hat{\theta}\|$ alone is a geometric metric, not a continuous-optimality certificate. However, under standard assumptions, a conservative window-specific Lipschitz constant L_Φ for the conditional CVaR would provide the error bound:

$$0 \leq \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta) - \max_{\theta \in \hat{\mathcal{U}}} \Phi_\tau(w, \theta) \leq L_\Phi \rho. \quad (35)$$

5.3 Derivation of the global Lipschitz constant

We derive a global Lipschitz constant for the function $\theta \mapsto \Phi_\tau(w, \theta)$. The conditional CVaR is given by the partial minimum:

$$\begin{aligned} \Phi_\tau(w, \theta) &= \min_{z \in [-M_L, M_L]} F(w, z, \theta), \\ \text{where } F(w, z, \theta) &= z + \frac{1}{\tau} \sum_{t=1}^T p_t(\theta) [\ell_t(w) - z]_+. \end{aligned} \quad (36)$$

where $\ell_t(w) = -x_t^\top w$. The mapping $\theta \mapsto F(w, z, \theta)$ is smooth. However, the partial minimum $\Phi_\tau(w, \theta)$ is generally nonsmooth due to the non-uniqueness of the VaR minimizer $z^*(\theta)$ and the kink in the positive part function. By Danskin's theorem and Clarke subdifferential calculus, the subgradients of $\Phi_\tau(w, \theta)$ with respect to θ are contained in the convex hull of the gradients of $F(w, z, \theta)$ evaluated at all optimal minimizers $z \in \arg \min F(w, z, \theta)$.

The gradient of F with respect to θ is:

$$\nabla_\theta F(w, z, \theta) = \frac{1}{\tau} \sum_{t=1}^T \nabla_\theta p_t(\theta) [\ell_t(w) - z]_+ \quad (37)$$

Because any VaR minimizer z lies in $[-M_L, M_L]$ and $\ell_t(w) \in [-M_L, M_L]$, the term $[\ell_t(w) - z]_+$ is uniformly bounded by $2M_L$. For Nadaraya–Watson weights with a Gaussian kernel $K_H(u) \propto \exp(-\frac{1}{2}u^\top H^{-1}u)$ one has $\nabla_\theta p_t(\theta) = p_t(\theta) H^{-1}(y_{t-1} - \bar{y}(\theta))$ with $\bar{y}(\theta) = \sum_s p_s(\theta) y_{s-1}$. Since $\bar{y}(\theta)$ is a convex combination of the sample states, $\|y_{t-1} - \bar{y}(\theta)\| \leq \|y_{t-1} - \theta\| + \max_s \|y_{s-1} - \theta\| \leq 2R$ with $R = \sup_{\theta, t} \|y_{t-1} - \theta\|$, and $\|H^{-1}\|_2 = 1/\lambda_{\min}(H)$, so that

$$\|\nabla_\theta p_t(\theta)\| \leq \frac{2R}{\lambda_{\min}(H)} p_t(\theta). \quad (38)$$

The factor $p_t(\theta)$ is essential: it is what makes the sum over t collapse, since $\sum_{t=1}^T p_t(\theta) = 1$ for every θ . Summing the T bounds therefore yields a constant that does not grow with the sample size:

$$L_\Phi := \frac{4RM_L}{\tau \lambda_{\min}(H)} \quad (39)$$

In the numerical implementation we use the Euclidean norm on the model state coordinates $(\log \text{VIX}, D)$. Over the aligned sample these coordinates span

$[2.21, 4.42] \times [0, 0.48]$, so the largest distance between a sample state and a point of the expanded domain is $R \approx 2.5$. The maximum absolute daily portfolio loss is $M_L \approx 0.23$, and the plug-in bandwidth $H = T^{-1/3} \hat{\Sigma}_y$ has $\lambda_{\min}(H) \approx 5.9 \times 10^{-5}$, the small eigenvalue reflecting the low variance of the drawdown coordinate relative to log-VIX. With $\tau = 0.05$ these values give $L_\Phi \approx 8 \times 10^5$. Combined with the grid-dispersion radius $\rho \approx 0.03$ of the 21×21 grid, the certified bound $L_\Phi \rho$ is several orders of magnitude larger than the objective values themselves, and is therefore vacuous in practice. This is precisely why the manuscript reports empirical local-search diagnostics rather than relying on the certified bound, and why tighter region-specific Lipschitz estimates are identified as future work.

5.4 A posteriori gap estimation via local search

While the adaptive algorithm operates on the finite grid $\hat{\mathcal{U}}$, we evaluate the approximation error over the continuous domain $\mathcal{U}$. If the minimizing VaR $z^*(\theta)$ is unique, Danskin's theorem yields the gradient with respect to θ . If the minimizer is non-unique, directional derivatives or generalized derivatives must be considered. Under uniqueness, the gradient is given by:

$$\nabla_\theta \Phi_\tau(w, \theta) = \frac{1}{\tau} \sum_{t=1}^T \nabla_\theta p_t(\theta) [\ell_t(w) - z^*(\theta)]_+, \quad (40)$$

where $\nabla_\theta p_t(\theta) = p_t(\theta) H^{-1} (y_{t-1} - \bar{y}(\theta))$ with $\bar{y}(\theta) = \sum_{s=1}^T p_s(\theta) y_{s-1}$. At nondifferentiability points, generalized directional derivatives or Clarke subgradients must be used. For bounding purposes, this permits the derivation of a computable uniform Lipschitz constant $L_\Phi := \frac{4RM_L}{\tau \lambda_{\min}(H)}$, where $R = \sup_{\theta, t} \|y_{t-1} - \theta\|$ is the maximum state distance and M_L bounds the portfolio loss. The grid oracle thus constitutes an inexact separation oracle with additive, one-sided inexactness $\delta \leq L_\Phi \rho$. We state the consequence directly: since the master lower bounds satisfy $\text{LB}_k \leq v^*$ for every $\hat{\mathcal{U}}_k \subseteq \mathcal{U}$, and since at termination $\max_{\theta \in \hat{\mathcal{U}}} \Phi_\tau(w^k, \theta) - \text{LB}_k \leq \varepsilon$, the incumbent satisfies

$$0 \leq G(w^k) - v^* \leq \varepsilon + L_\Phi \rho. \quad (41)$$

This additive-oracle propagation complements the analysis of [25], whose relative-gap δ -oracle model (and, for their cutting-plane rate results, a strong-convexity assumption on the objective) does not directly subsume our linear master objective. We validate the analytical gradient $\nabla_\theta \Phi_\tau$ using central finite differences, confirming numerical agreement to within 10^{-7} precision. A direct evaluation of the conservative bound $\frac{4RM_L}{\tau \lambda_{\min}(H)}$ with the sample values reported in Section 5.3 yields $L_\Phi \approx 8 \times 10^5$, which is far too loose to be practically informative. We therefore rely on a heuristic a posteriori gap estimation. Column two of Table 2 reports the maximum observed gradient norm over a dense 201×201 reference grid (which provides a local empirical gauge of the function's steepness, though not a formal global Lipschitz constant). Column five reports the empirically observed gap $\tilde{G}(w^*) - \hat{G}(w^*)$, where $\tilde{G}(w^*)$ is the best value found by refined-grid evaluation followed by projected gradient ascent (five starts from the top grid maximizers, box-projected Armijo steps, stopping at projected gradient mapping norm $< 10^{-8}$). Across all tested windows, the observed gap $\tilde{G}(w^*) - \hat{G}(w^*)$ is below 10^{-4} (column

five of Table 2). Because $\tilde{G}$ relies on local search, it strictly lower-bounds the true continuous supremum. Therefore, the true discretization gap could theoretically exceed this observed gap. Nevertheless, the small values provide strong heuristic confidence in the practical adequacy of the 21×21 grid. (In the COVID-19 window, the local search improvement is identically zero because the projected gradient ascent terminated without escaping the initial grid-node maximizer.)

Table 2 A posteriori gap estimation for the robust discretization

Crisis Period	Max Gradient Norm	Grid Dispersion	Gradient $\times$ Dispersion	Local Search Improvement
1998-08 (LTCM)	13.743	0.043	0.59	7.04e-04
2008-10 (Lehman)	22.105	0.064	1.41	6.94e-18
2013-05 (Taper Tantrum)	5.451	0.061	0.33	1.39e-17
2020-03 (COVID-19)	96.477	0.067	6.46	0
2022-09 (Inflation Shock)	13.712	0.067	0.92	1.39e-17

5.5 Dense reference-grid validation of the discretization

The certified bound $L_{\Phi}\rho$ of Section 5.3 is too loose to be informative, and the local search of Table 2 is heuristic. We therefore also evaluate the separation objective directly on a dense reference grid. Because $\theta \mapsto \Phi_{\tau}(w^k, \theta)$ costs $O(T)$ per evaluation on a two-dimensional box, a 1001×1001 grid of approximately 10^6 candidate states is computationally routine.

Across 20 evenly spaced rolling windows, the largest excess of the dense-grid maximum over the value returned by the 21×21 separation oracle is

$$\max_k \left[\max_{\theta \in \tilde{\mathcal{U}}^{1001}} \Phi_{\tau}(w^k, \theta) - \max_{\theta \in \tilde{\mathcal{U}}^{21}} \Phi_{\tau}(w^k, \theta) \right] = 8.53 \times 10^{-4},$$

that is 0.085 percentage points, against worst-case CVaR levels of approximately 4.5%. The discretization error of the operational grid is therefore below 2% in relative terms on every window tested. This remains an empirical bound rather than a certificate, since the dense grid is itself finite; but at 10^6 points its dispersion radius is roughly fifty times smaller than that of the operational grid, so it provides a far more demanding check than the local search alone.

6 Experimental design

We evaluate the out-of-sample performance, turnover efficiency, downside capital preservation, and computational scalability of the continuous-state robust portfolio framework.

6.1 Data and rolling-window backtesting protocol

The empirical universe comprises the Kenneth French 30 Industry Portfolios [16] (value-weighted daily returns), providing broad industry-level coverage of the US equity market. Daily asset returns are aligned with:

- **CBOE VIX**: Daily closing prices of the CBOE Volatility Index.

- **Broad Market Drawdown:** Calculated daily from the equal-weighted 30-industry composite (chosen for its heightened sensitivity to broad-based sector distress) over trailing 63-day lookback windows ($D_t \in [0, 1]$).

The aligned daily dataset spans from January 1990 to 30 June 2026, comprising 9,186 daily observations.

To clarify the chronology of the 377 rolling evaluation periods over the approximately 31.5-year span, Table 3 provides the exact backtest calendar. The first 1,260 days strictly constitute the initial training window. The first allocation is computed at the close of 27 December 1994. The terminal 9 days of the dataset are omitted to maintain equally sized 21-day compounding periods.

Table 3 Exact chronological boundaries of the rolling out-of-sample backtest

Chronological Marker	Date / Value
End of first estimation window	27 December 1994
First rebalance decision date	27 December 1994
First out-of-sample daily return date	28 December 1994
End of first 21-day holding period	26 January 1995
Last rebalance decision date	19 May 2026
Last out-of-sample return date (compounded)	17 June 2026
Number of full 21-day periods	377
Unused terminal partial period	9 trading days (until 30 June 2026)

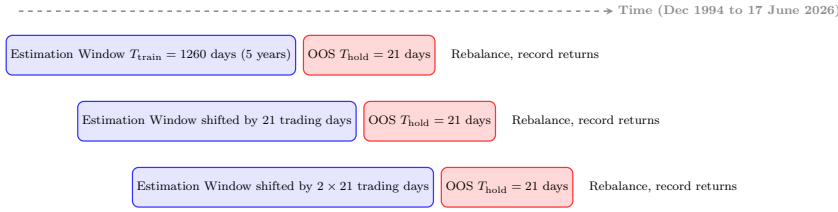


Fig. 3 The rolling-window backtesting protocol

The backtest uses a rolling estimation window of $T_{\text{train}} = 1260$ trading days (approximately 5 years), as illustrated in Figure 3. At each rebalancing date, the optimization models are estimated using only data available up to that point. The resulting allocations are held constant over the subsequent out-of-sample holding period of $T_{\text{hold}} = 21$ trading days. The backtest generates 377 consecutive 21-trading-day holding periods from 28 December 1994 to 17 June 2026.

To improve comparability across the four optimized models, the target expected return μ_{target} is dynamically set at each rebalancing date to the cross-sectional median of historical asset means: $\mu_{\text{target}} = \text{median}(\hat{\mu})$. The identical target return constraint $\hat{\mu}^\top w \geq \mu_{\text{target}}$ and weight cap $\bar{w} = 0.15$ are enforced on Robust SIP, Nominal CVaR, and Finite-Regime CVaR. In the computational implementation, μ_{target} is dynamically clamped to the attainable interval $[\mu_{\min}, \mu_{\max} - 10^{-6}]$ to guarantee strict feasibility. Over the 377 rolling out-of-sample evaluation windows, target return clamping occurred in exactly 0 periods, ensuring the original target specifications were strictly respected throughout the backtest.

We compare the continuous-state Robust SIP against four established benchmark strategies:

- **1/N:** Naive equal-weighted portfolio $w_i = 1/N$.

- **Target-Constrained Minimum Variance (TC-MinVar)**: Minimizes portfolio variance $w^\top \hat{\Sigma} w$ subject to $w \in W$, using the standard sample covariance matrix $\hat{\Sigma}$. Solved via quadratic programming.
- **Nominal CVaR**: Minimizes unconditional historical CVaR at 95% confidence level subject to $w \in W$, using uniform observation weights $1/T$.
- **Finite-Regime CVaR**: Minimizes the maximum CVaR across $K = 4$ discrete regimes. Regimes are formed by independent median splits on log-VIX and trailing drawdown. Regime probabilities are uniform over the observations within each regime, and boundaries are recomputed in every rolling window.

To incorporate realistic market frictions, we compute holding-period portfolio turnover taking into account asset price drift over the holding period:

$$\tilde{w}_{i,t} = \frac{w_{i,t-1}(1 + R_{i,t})}{\sum_{j=1}^N w_{j,t-1}(1 + R_{j,t})} \quad (42)$$

where $R_{i,t}$ is the cumulative simple/net return (and $1 + R_{i,t}$ is the gross return) of asset i over the 21-day holding period. Holding-period turnover and net return are defined as:

$$\text{TO}_t = \frac{1}{2} \sum_{i=1}^N |w_{i,t} - \tilde{w}_{i,t}| \quad (43)$$

and

$$r_t^{\text{net}} = r_t^{\text{pre-cost}} - c \text{TO}_t, \quad (44)$$

where $r_t^{\text{pre-cost}}$ is the portfolio's simple holding-period return before transaction costs.

7 Main empirical results

The optimization objective is based on one-day conditional returns, whereas portfolio weights are held for 21 trading days. This mismatch is intentional. Daily observations provide a substantially larger effective sample for estimating state-dependent tail distributions, since the conditional CVaR at a given state is estimated from kernel-weighted daily losses; aggregating to the holding frequency before estimation would divide the usable sample by roughly twenty and leave the tail of each conditional distribution determined by a handful of observations. Monthly rebalancing, on the other hand, keeps turnover and transaction costs at implementable levels. The model therefore uses high-frequency information for estimation and lower-frequency implementation for the portfolio decision, and all reported performance statistics are computed from the realized 21-day holding-period returns.

7.1 Long-term cumulative wealth and risk-adjusted performance

Figure 4 displays the cumulative wealth trajectories of all five investment strategies net of 10 bps transaction costs on an initial \$1 investment. The naive $1/N$ portfolio delivers the highest final cumulative wealth of \$29.25, followed by the Finite-Regime CVaR portfolio (\$25.68), the continuous Robust SIP strategy (\$25.04), the Nominal CVaR portfolio (\$22.92), and the TC-MinVar portfolio (\$22.65).

Table 4 reports the 5-metric performance profile. Key findings include:

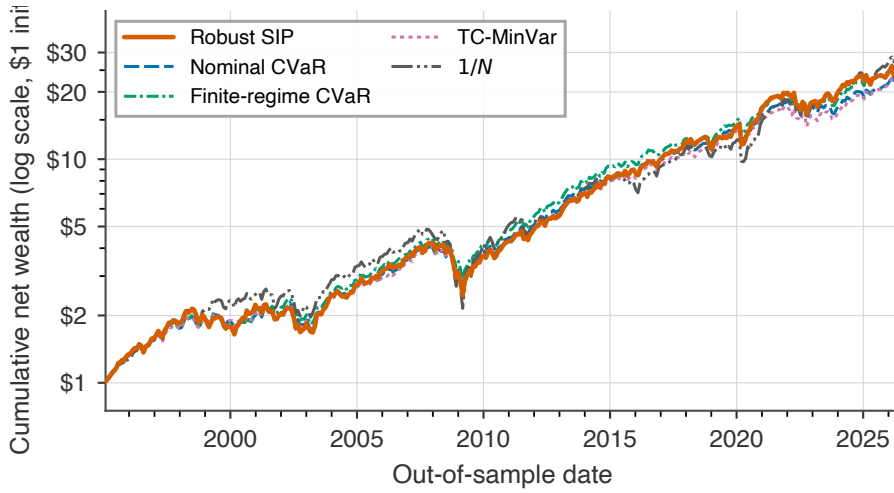


Fig. 4 Out-of-sample cumulative net wealth trajectory

Table 4 Comprehensive out-of-sample performance summary (Sharpe ratios assume a zero risk-free rate)

Strategy	Return (%)	Vol (%)	Sharpe	Max DD (%)	Turnover (%)
1/N	12.24	17.00	0.720	-55.79	1.69
TC-MinVar	10.73	12.26	0.875	-40.18	5.93
Nominal CVaR	10.76	12.23	0.880	-35.76	8.23
Finite-Regime CVaR	11.15	12.38	0.901	-36.76	8.78
Robust SIP	11.28	13.95	0.809	-41.69	9.48

1. **Return Generation:** Robust SIP achieves an annualized mean return of 11.28%, exceeding the point estimates of Nominal CVaR (10.76%), TC-MinVar (10.73%), and Finite-Regime CVaR (11.15%). It ranks second overall, behind the naive 1/N portfolio (12.24%). The 1/N portfolio benefits from broad cross-sectional diversification and full exposure to the long-run equity market risk premium over the 31-year evaluation period.
2. **Risk-Adjusted Ratios:** Finite-Regime CVaR achieves the highest Sharpe ratio (0.901), followed by Nominal CVaR (0.880) and TC-MinVar (0.875). Robust SIP records a Sharpe ratio of 0.809. The resulting allocations exhibit higher realized volatility and greater concentration. Determining whether this arises from return targeting, state conditioning, or sparse-boundary effects requires further decomposition.
3. **Turnover and Concentration:** Robust SIP generates the highest average holding-period turnover (9.48%) among optimized strategies, resulting from rolling-window re-estimation of the return-state relationship. The resulting approximate transaction cost drag is 11.37 basis points per year. The effective number of assets ($N_{\text{eff}} = 7.36$) is the most concentrated, reflecting the concentrated allocations produced by worst-state CVaR minimization under the rolling state domain and return-target constraint.

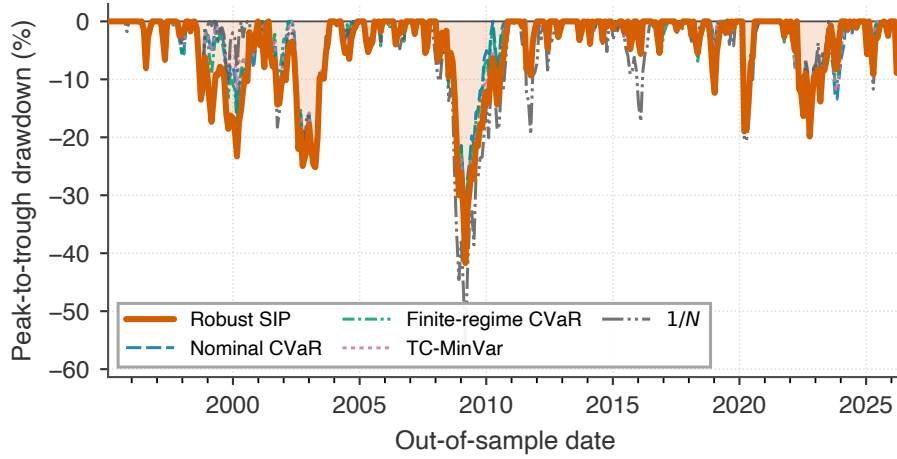


Fig. 5 Out-of-sample portfolio drawdowns over time across all strategies

7.2 Drawdown profiles and crisis-period performance

Figure 5 plots the time series of out-of-sample drawdowns. The naive $1/N$ strategy experiences severe capital destruction during major market dislocations, suffering a maximum peak-to-trough drawdown of -55.79% during the 2008 financial crisis. All optimized models experienced smaller realized maximum drawdowns than the $1/N$ portfolio over the reported sample: Nominal CVaR achieves the shallowest at -35.76% , Finite-Regime CVaR reaches -36.76% , and TC-MinVar reaches -40.18% . Robust SIP records a maximum drawdown of -41.70% , which is deeper than the unconditional CVaR benchmarks.

Table 5 Crisis-period performance. Crisis intervals are defined ex ante by the dates shown and applied identically to all strategies

Crisis period	Window	Strategy	Return (%)	Max DD (%)
DotCom	1 Mar 2000–31 Oct 2002	1/N	-4.75	-24.41
DotCom	1 Mar 2000–31 Oct 2002	TC-MinVar	1.36	-20.66
DotCom	1 Mar 2000–31 Oct 2002	Nominal CVaR	5.86	-20.66
DotCom	1 Mar 2000–31 Oct 2002	Finite-Regime CVaR	17.26	-20.33
DotCom	1 Mar 2000–31 Oct 2002	Robust SIP	4.25	-24.95
GFC	1 Oct 2007–31 Mar 2009	1/N	-53.56	-55.79
GFC	1 Oct 2007–31 Mar 2009	TC-MinVar	-34.67	-40.18
GFC	1 Oct 2007–31 Mar 2009	Nominal CVaR	-31.13	-35.76
GFC	1 Oct 2007–31 Mar 2009	Finite-Regime CVaR	-30.91	-36.76
GFC	1 Oct 2007–31 Mar 2009	Robust SIP	-36.49	-41.69
COVID	1 Feb 2020–30 Apr 2020	1/N	-20.18	-20.64
COVID	1 Feb 2020–30 Apr 2020	TC-MinVar	-12.25	-15.29
COVID	1 Feb 2020–30 Apr 2020	Nominal CVaR	-12.05	-15.42
COVID	1 Feb 2020–30 Apr 2020	Finite-Regime CVaR	-11.67	-15.15
COVID	1 Feb 2020–30 Apr 2020	Robust SIP	-17.44	-18.99
Inflation	1 Jan 2022–31 Dec 2022	1/N	-6.05	-17.24
Inflation	1 Jan 2022–31 Dec 2022	TC-MinVar	-6.47	-18.12
Inflation	1 Jan 2022–31 Dec 2022	Nominal CVaR	-4.83	-17.92
Inflation	1 Jan 2022–31 Dec 2022	Finite-Regime CVaR	-7.66	-18.88
Inflation	1 Jan 2022–31 Dec 2022	Robust SIP	-8.59	-19.86

Table 5 presents the crisis-period analysis. During the Dot-Com crash, Finite-Regime CVaR achieved the highest cumulative return ($+17.26\%$), followed by Nominal CVaR ($+5.86\%$) and Robust SIP ($+4.25\%$), while $1/N$ suffered the most (-4.75%). In the Global Financial Crisis, Nominal CVaR provided the best downside

protection (-31.13% cumulative, -35.76% maximum drawdown), while Robust SIP experienced a deeper drawdown of -41.69% . During the COVID-19 crash, all optimized models limited losses to the -11% to -17% range, with Robust SIP absorbing a larger loss (-17.44%) than the unconditional benchmarks (-12%). In the 2022 inflation-driven selloff, Nominal CVaR was most resilient (-4.83%), while Robust SIP suffered a larger cumulative loss (-8.59%). These results show that continuous-state robustification does not uniformly improve realized crisis-period performance. Rather, it reshapes the allocation profile in ways that can amplify exposure during acute market dislocations.

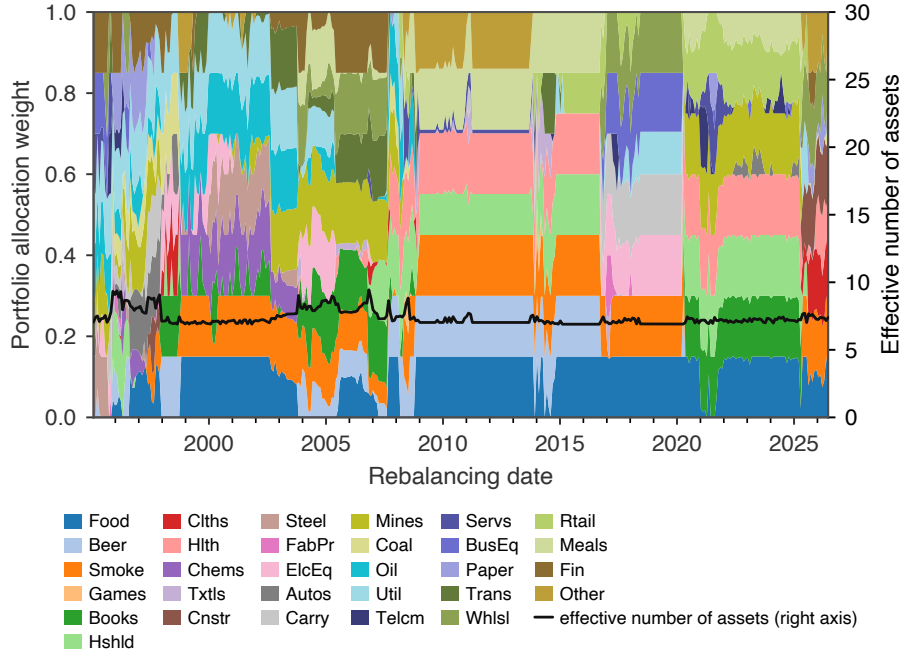


Fig. 6 Robust SIP industry allocations over time. Stacked areas are portfolio weights across the 30 industries; the black line is the effective number of assets, read on the right-hand axis

Figures 6 and 7 compare the dynamic industry allocations of Robust SIP and TC-MinVar, together with the evolution of portfolio concentration measured by the effective number of assets. Robust SIP maintains allocations across 7 to 10 active industry sectors ($N_{\text{eff}} = 7.36$), reallocating capital across cyclical and defensive sectors as historical window dynamics evolve. Conversely, the TC-MinVar portfolio remains concentrated in low-beta industries ($N_{\text{eff}} = 8.15$), and exhibits lower realized volatility and different sector exposures.

Figure 8 confirms that the turnover distribution for Robust SIP has a mean near 9.5% , reflecting active rebalancing induced by rolling re-estimation of the return-state relationship, kernel weights, state-domain boundaries, and return target. TC-MinVar exhibits lower mean turnover (5.93%), while $1/N$ has the lowest (1.7%) due to its static equal-weight target. Figure 9 accumulates these per-rebalancing differences into the total transaction-cost drag borne by each strategy over the out-of-sample period.

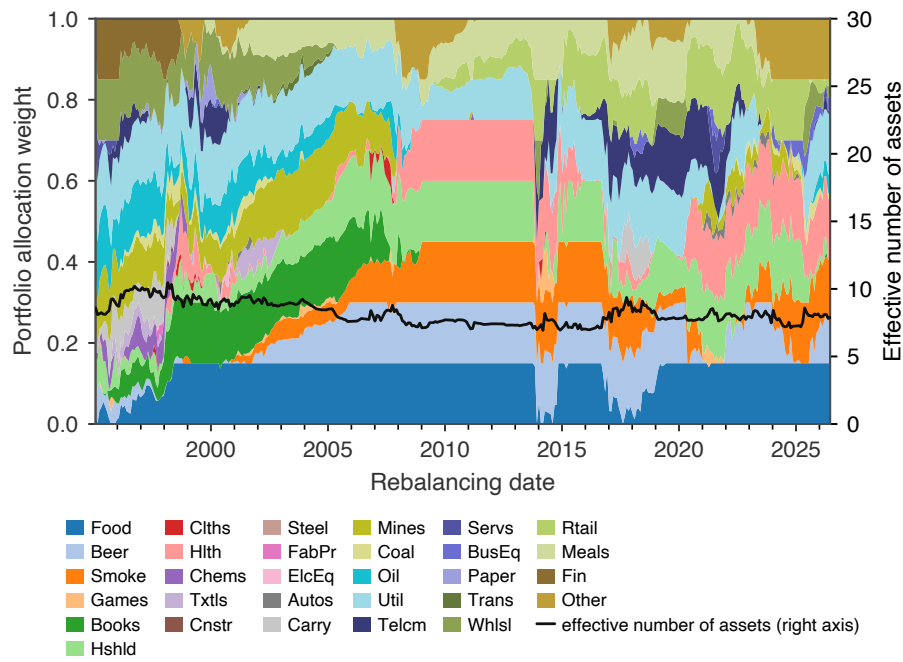


Fig. 7 TC-MinVar industry allocations over time. Stacked areas are portfolio weights across the 30 industries; the black line is the effective number of assets, read on the right-hand axis

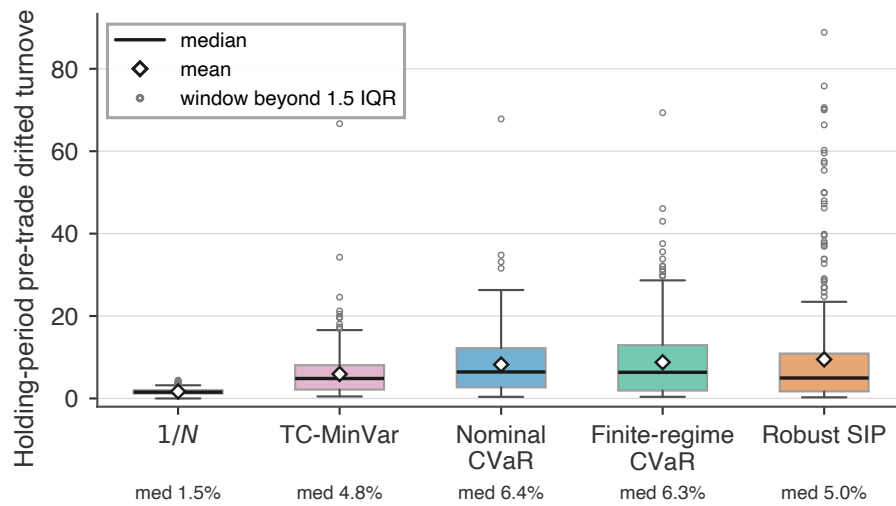


Fig. 8 Distribution of holding-period portfolio turnover

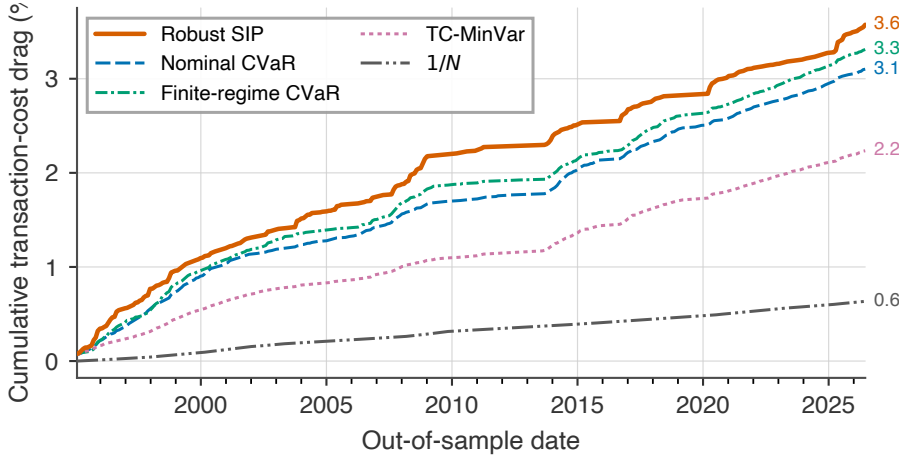


Fig. 9 Cumulative transaction cost impact

Table 6 Transaction cost sensitivity: out-of-sample performance under alternative cost levels

Strategy	Sharpe Ratio (net of TC)					Final Wealth (\$, net of TC)				
	0 bps	5 bps	10 bps	20 bps	50 bps	0 bps	5 bps	10 bps	20 bps	50 bps
1/N	0.721	0.720	0.720	0.719	0.715	29.43	29.34	29.25	29.06	28.52
TC-MinVar	0.881	0.878	0.875	0.869	0.852	23.16	22.90	22.65	22.16	20.73
Nominal CVaR	0.888	0.884	0.880	0.872	0.848	23.63	23.27	22.92	22.22	20.26
Finite-Regime CVaR	0.909	0.905	0.901	0.892	0.867	26.53	26.10	25.68	24.85	22.52
Robust SIP	0.817	0.813	0.809	0.801	0.777	25.94	25.49	25.04	24.17	21.74

Table 6 examines transaction cost sensitivity across 0, 5, 10, 20, and 50 bps. Finite-Regime CVaR records the highest point-estimate Sharpe ratio across all reported transaction-cost levels, maintaining a Sharpe ratio above 0.867 even at 50 bps. Robust SIP remains competitive in absolute wealth generation: at 50 bps, its final wealth of \$21.74 exceeds Nominal CVaR (\$20.26) and TC-MinVar (\$20.73), though it trails Finite-Regime (\$22.52). The 1/N portfolio is least sensitive to transaction costs due to its minimal turnover, retaining \$28.52 at 50 bps.

7.3 What the empirical evidence does and does not support

It is worth stating plainly what Tables 4 and 6 do not show. Robust SIP does not record the highest Sharpe ratio: at zero transaction cost it reaches 0.809, behind Finite-Regime CVaR (0.901), Nominal CVaR (0.880) and TC-MinVar (0.875), and Section 8 will show that differences of this magnitude cannot be distinguished from zero by a block bootstrap. Nor does it dominate the CVaR benchmarks during the principal crisis episodes. Any reading of this study as a demonstration that continuous-state robustification improves risk-adjusted performance would misstate the evidence. Section 9.5 shows that this figure is itself sensitive to the return-target convention, rising to 0.899 within the Robust SIP specification under a tighter target. That experiment holds the benchmarks at their own baseline target and therefore does not license a cross-strategy comparison at the alternative levels; it establishes only that the return-target convention accounts for a material part of the performance profile reported here.

The empirical contribution lies elsewhere, and the relevant comparison is not against the unconditional benchmarks but against the alternative way of using the same state information. Conditioning the return distribution directly on the observed state, without a robust layer, turns over 35.7% of the portfolio per rebalancing and ends with \$17.82, against 9.48% and \$25.04 for the robust formulation (Section 9.6); Section 9.7 shows that shrinking the conditional law towards the unconditional one does not recover the difference at any intermediate degree of shrinkage. Robustness over a state domain therefore acts as a regularization mechanism against excessive responsiveness to short-lived movements in the conditioning variables: it requires satisfactory tail-risk performance over an entire set of plausible states rather than tracking the one state currently observed, and it delivers state-awareness at close to unconditional trading cost. That, together with the computational sparsity documented in Section 8.1, is what the experiment supports.

8 Computational results

Figure 10 illustrates the number of active stress states retained by the master problem across all 377 rolling backtest windows. Across the entire 31-year backtest, the algorithm consistently converges while retaining between 1 and 9 active market states (average 3.66) out of the 441 candidate states in $\hat{\mathcal{U}}$. Under the implemented initialization and one-state-per-iteration update rule, the reported number of active states coincides with the recorded number of master-problem solves (average 3.66 iterations).

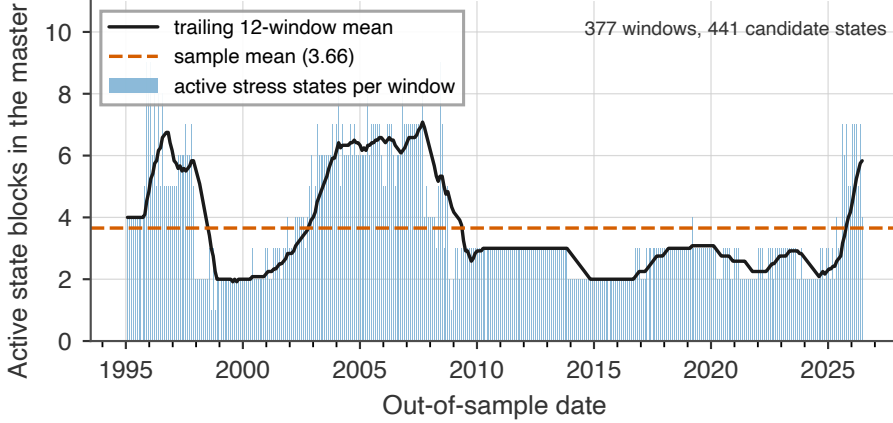


Fig. 10 Active state blocks across rolling windows

Figure 11 tracks the convergence of the Master LP Lower Bound (LB_k) and the Grid Separation Worst-Case Value ($\hat{G}_k$). The shaded green area denotes the grid-restricted exchange gap ($\hat{G}_k - LB_k$). Starting from an initial gap at iteration $k = 1$, the oracle worst-case value fluctuates as the portfolio changes, while the master lower bound increases monotonically and the residual gap eventually closes. After 3 exchange iterations, the grid-restricted residual gap falls below 10^{-4} , confirming numerical convergence to the finite robust problem over the 21×21 candidate grid.

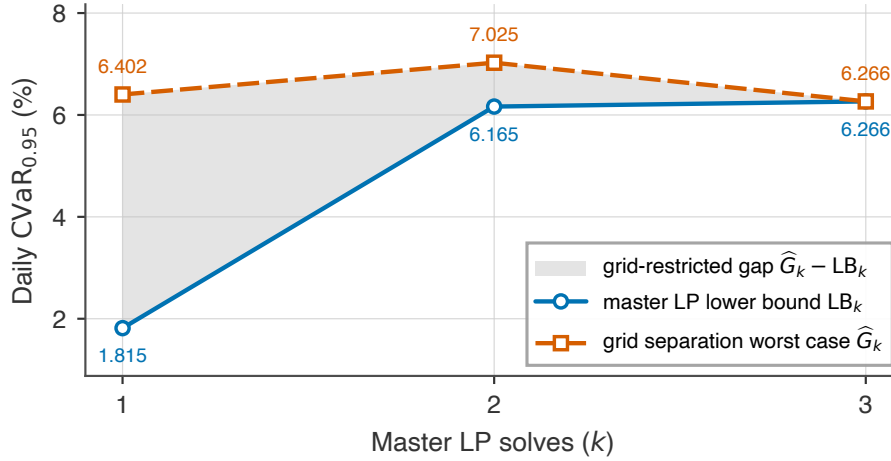


Fig. 11 Grid-restricted convergence of the lower and upper bounds

Table 7 Computational benchmark: adaptive exchange algorithm vs. dense grid approximation

Method	Status	State blocks	Runtime	Reported value
Adaptive SIP 21×21	Exchange tolerance reached	5	3.45 s	final grid worst-case CVaR $\approx 3.3585\%$
Dense 21×21	OPTIMAL	441	71.43 s	optimum $\approx 3.3578\%$
Dense 41×41	TIME_LIMIT / no primal solution	1681	632.25 s	N/A

Table 8 Distribution of the window-level mean ESS across active stress states

Statistic	Value
Minimum window-level mean ESS	2.18
5th Percentile window-level mean ESS	17.0
Median window-level mean ESS	72.8
Mean window-level mean ESS	83.0
Maximum window-level mean ESS	264.4

Figure 11 illustrates convergence for one selected rolling window. Table 7 reports a single representative-window snapshot used specifically for the dense-grid benchmarking, as running the exact dense 41×41 LP over all 377 windows proved computationally prohibitive. Therefore, the objective values and numbers of active states in Figure 11 and Table 7 are not expected to coincide. On that window the adaptive algorithm reached the exchange tolerance in 3.45 seconds against 71.43 seconds for the dense 21×21 extensive-form LP, a 20.7-fold speedup, and the dense 41×41 LP did not return a primal solution within the 600-second solver limit. (We note that a row-generation implementation on the dense grid would provide a more competitive alternative than the extensive-form LP.)

Several diagnostics in Tables 7 and 8 warrant careful interpretation:

1. **Representative-Window ℓ_1 Weight Discrepancy:** On the representative window, the ℓ_1 distance between the adaptive 21×21 exchange solution and the dense 21×21 grid solution is $\|w_{\text{adaptive}} - w_{\text{dense21}}\|_1 \approx 0.00156$. The difference between the adaptive final grid worst-case value and the dense-grid optimum is approximately 7.6×10^{-6} , below the exchange tolerance of 10^{-4} .
2. **Active-State Support:** The minimum, across rolling windows, of the within-window mean ESS over active states is 2.18, indicating that some windows

contain collectively low-support active states. The reported value $\tau \times \text{ESS}$ is only a descriptive tail-support proxy and is not a formal tail effective sample size.

Figure 12 traces the window-level mean ESS over active stress states across the 377 rolling windows, showing that the low-support episodes are concentrated in a small number of windows rather than spread uniformly over the sample.

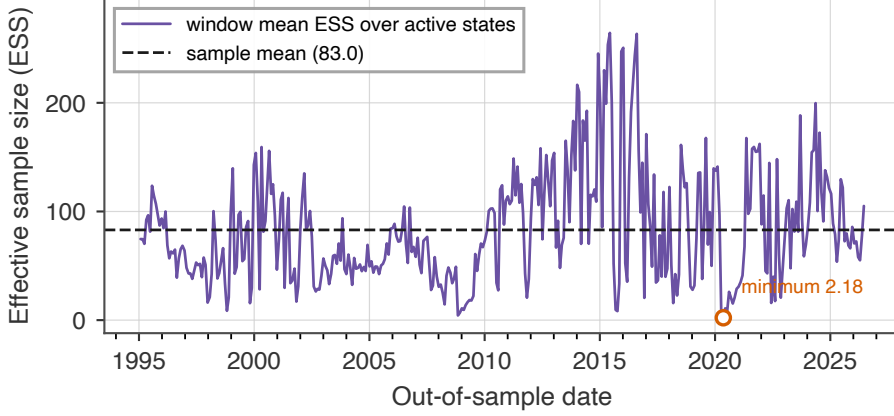


Fig. 12 Evolution of the average Effective Sample Size

8.1 Scalability in the grid dimension

Table 7 compares the two methods on a single window, which is a weak basis for a computational claim. We therefore repeated the comparison on 40 rolling windows, chosen to span the sample evenly while over-representing the highest log-VIX and deepest drawdown states, and solved every window twice: once with the extensive-form dense LP over the full candidate grid, and once with the adaptive exchange algorithm. Both solvers received identical data, identical estimation settings and identical grids. Table 9 reports the outcome for three grid resolutions.

Table 9 Matched dense-LP and exchange-algorithm comparison over 40 rolling windows per grid resolution

Grid	States	Dense LP (s)	Exch. (s)	Speed-up	Active blocks	Max gap	Mean ℓ_1
11×11	121	6.16	0.235	36.2	3.33	7.5×10^{-7}	0.00001
21×21	441	27.06	0.316	110.3	3.27	1.2×10^{-5}	0.00018
31×31	961	105.35	0.813	242.1	3.42	2.0×10^{-5}	0.00017

Three observations follow. First, the two methods agree: the largest absolute difference between the dense optimum and the exchange value over all 120 solves is 2.0×10^{-5} , an order of magnitude below the exchange tolerance of 10^{-4} , and the mean ℓ_1 distance between the two allocations never exceeds 0.0002. The exchange algorithm is therefore solving the same problem, not an easier one. Second, the speedup is not an artifact of one instance: the paired per-window speedup is at

least 11.3 on every one of the 120 solves, and its median rises from 33.1 to 234.3 as the grid is refined. Third, and most importantly, the number of active state blocks does not grow with the grid. As the candidate set expands by a factor of 7.9, from 121 to 961 states, the mean number of blocks retained by the master problem moves only between 3.27 and 3.42, and never exceeds 6 in any window. Dense solution time grows by a factor of 17.1 over the same refinement while exchange time grows by a factor of 3.5.

The computational contribution is therefore not the speedup measured on one benchmark instance, but this stability: the binding subset of the state space stays small as the discretization is refined, so the cost of the master problem is governed by the number of states that actually bind rather than by the resolution of the candidate grid. This is the property that makes a fine discretization of a continuous state space affordable, and it is what a purely dense formulation cannot offer. We did not extend the matched comparison beyond 31×31 because the extensive-form dense LP becomes the binding constraint, not the exchange algorithm; the single-window result in Table 7, where the dense 41×41 LP returned no primal solution within 600 seconds, illustrates the point.

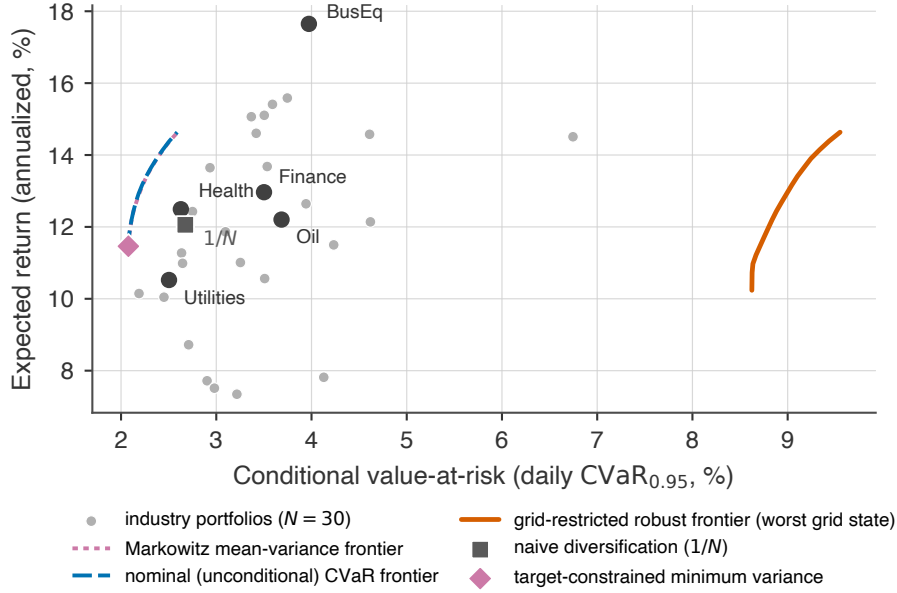


Fig. 13 In-sample risk-return efficient frontiers (15% weight cap)

Figure 13 contextualizes the in-sample decision environment. The grid-restricted robust efficient frontier forms an outer envelope of worst-grid-state CVaR values, positioned to the right of the unconditional nominal CVaR frontier and the classical Markowitz mean-variance frontier. Individual industry portfolios exhibit substantial dispersion in both expected return and downside tail risk, with defensive sectors displaying lower tail risk compared to high-beta sectors.

Figure 14 visualizes the continuous market-state space $\mathcal{U} \subset \mathbb{R}^2$ alongside the bivariate kernel density estimate $f(\log\text{-VIX}, \text{Drawdown})$. Historical market states concentrate in a low-volatility, low-drawdown core regime (VIX between 10 and 20, Drawdown below 10%). However, the continuous domain extends outward

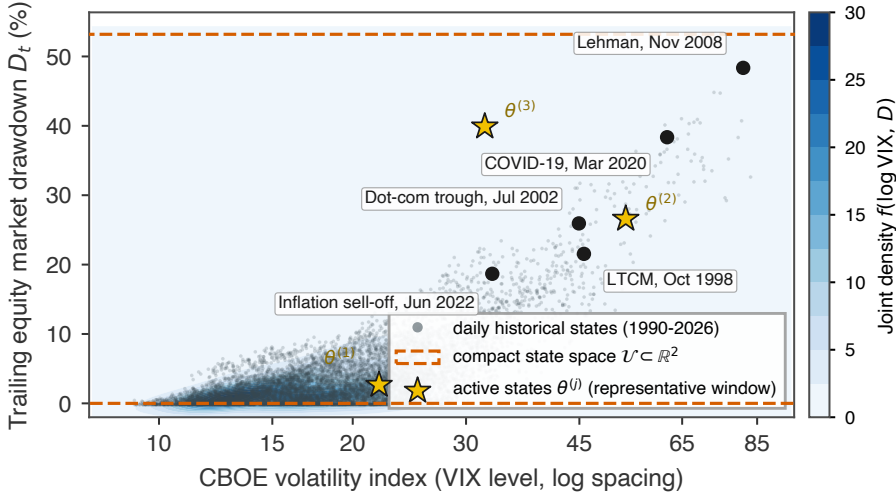


Fig. 14 State density and active stress states

into severe tail regions, encompassing major historical dislocations such as the 1998 LTCM collapse (VIX 45.74, Drawdown 19.3%), the 2000 to 2002 Dot-Com crash (VIX 45.08, Drawdown 44.7%), the 2008 Lehman Brothers crisis (VIX 80.06, Drawdown 48.4%), the March 2020 COVID-19 shock (VIX 82.69, Drawdown 33.8%), and the 2022 inflation-driven bear market (VIX 34.02, Drawdown 24.5%). In the representative visualization, some active stress states lie in relatively low-density regions of the state domain, exerting binding constraints on the master portfolio problem. To mitigate the computational burden and prevent over-conservatism, one could restrict the Cartesian grid $\hat{U}$ to the empirical convex hull of the observed data, thereby eliminating artificial corner states that lack any historical precedent while preserving the interpolative properties of the kernel density.

To rigorously assess whether the observed performance differences are statistically distinguishable from sample noise, we implement the circular block percentile bootstrap of Politis and Romano [26] for the Sharpe-ratio difference. Because financial returns exhibit temporal dependence and volatility clustering, standard i.i.d. t -tests produce severely understated standard errors. This nonparametric inference provides an exploratory assessment of performance stability. We resample holding-period strategy return pairs using a block size of $b = 12$ periods over $B = 5,000$ bootstrap replications. Table 10 reports the point estimate, two-sided 95% confidence interval, and bootstrap p -value for the zero-rate Sharpe ratio difference $\Delta SR = SR_{\text{Robust}} - SR_{\text{Benchmark}}$ against all four benchmarks. The bootstrap p -values exceed 0.13 across all benchmarks, indicating that the null hypothesis of equal Sharpe ratios cannot be rejected at conventional significance levels at the 5% level.

Against all four benchmarks, the bootstrap p -values exceed 0.05 ($p = 0.281$ vs. Nominal CVaR, $p = 0.330$ vs. $1/N$, $p = 0.327$ vs. TC-MinVar, and $p = 0.130$ vs. Finite-Regime). The block-bootstrap analysis therefore provides no evidence that the observed Sharpe-ratio differences are statistically distinguishable from zero over the evaluation sample. This is an absence of evidence for a difference, not evidence that the strategies perform equally. As shown in Figure 15, the bootstrap

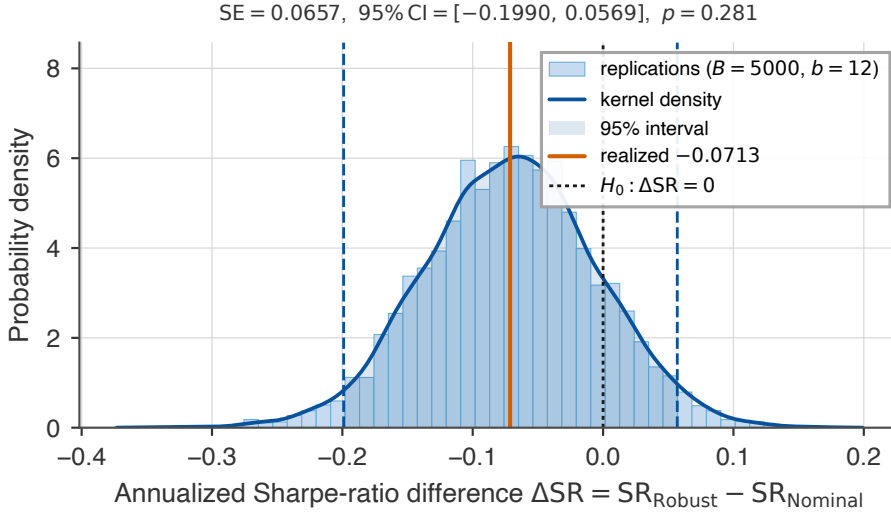


Fig. 15 Bootstrap distribution of the Sharpe-ratio difference (computed using a zero risk-free rate)

Table 10 Paired circular moving-block bootstrap inference for out-of-sample Sharpe ratio differences (zero risk-free rate)

Benchmark	Δ Sharpe	Std Error	95% CI	p -value
Nominal CVaR	-0.0713	0.0657	[-0.1990, 0.0569]	0.281
1/N	0.0890	0.0928	[-0.1036, 0.2584]	0.330
TC-MinVar	-0.0663	0.0674	[-0.2082, 0.0564]	0.327
Finite-Regime CVaR	-0.0919	0.0613	[-0.2157, 0.0261]	0.130

distribution of ΔSR relative to Nominal CVaR is approximately symmetric and centered near the observed estimate -0.071 . The bootstrap confidence intervals provide an exploratory gauge of estimation uncertainty, with the null value $\Delta SR = 0$ comfortably contained within the 95% confidence interval in all four comparisons.

This statistical finding provides an important nuance to the wealth accumulation results: while Robust SIP generates competitive cumulative wealth (\$25.04 vs. \$22.92 for Nominal CVaR), its risk-adjusted Sharpe ratio is lower due to elevated portfolio volatility. The bootstrap does not provide sufficient evidence to distinguish these Sharpe-ratio differences from zero at conventional significance levels. The primary methodological contribution of the continuous-state framework lies in its principled representation of the state-risk relationship and sparse constraint generation, rather than in statistically dominant out-of-sample risk-adjusted performance.

8.2 Reproducibility, open science, and codebase

Two distinct tolerances appear in this work and should not be confused. The solver feasibility and optimality tolerance 10^{-8} is internal to each individual LP or QP solve. The exchange termination tolerance $\varepsilon = 10^{-4}$ governs when the outer constraint-generation loop stops, that is when the grid-restricted gap $\hat{G}_k - LB_k$ is

deemed closed. The former controls the accuracy of a single subproblem; the latter controls the accuracy of the overall grid-restricted optimum.

To ensure scientific transparency and enable independent verification of all empirical results, the entire computational framework is open-source under an MIT License. The exact code and results corresponding to this manuscript are available in the official release [8].

The repository provides the complete pipeline structured into four main components:

1. **Optimization Engine** (`code/RobustSIP.jl`): Implements the grid-restricted adaptive constraint-generation algorithm, the HiGHS master-LP solver, the full-grid separation oracle, and the benchmark solvers.
2. **Backtesting Suite** (`code/main_exp.jl`): Executes the 31-year rolling-window out-of-sample backtest, computes the 14 performance metrics, performs transaction cost sensitivity sweeps, and conducts the paired circular moving-block bootstrap test.
3. **Empirical Data Module** (`data/aligned_market_data.csv`): Contains the synchronized daily returns for the Kenneth French 30 Industry Portfolios [16] alongside CBOE VIX and drawdown indicators over the full 1990 to 2026 period.
4. **Figure Generation Routine** (`code/generate_publication_figures.py`): Reproduces all analytical figures in high-resolution vector PDF format.

Computations were performed on an Apple M1 Pro with 8 CPU cores and 16 GB RAM, using Julia 1.11.6 [5], CSV 0.10.16, DataFrames 1.8.2, JuMP 1.31.1 [13], and HiGHS 1.24.1 [19] (with default presolve, primal and dual feasibility tolerances at 10^{-8} , and single-threaded execution; warm starts across exchange iterations were disabled to ensure strict benchmarking consistency). Reported solve times encompass model construction and solver execution.

9 Model sensitivity

To ensure the robustness of our findings to methodological hyperparameters, we conduct sensitivity analyses across key design choices: grid resolution, kernel bandwidth, Effective Sample Size (ESS) regularization, and bootstrap block length. The grid resolution and bandwidth analyses (Tables 11 and 12) report averages over a representative subset of 20 evenly spaced rolling out-of-sample windows spanning the full evaluation period from 28 December 1994 to 17 June 2026. This uniform sampling guarantees that the analyses span multiple market regimes, including crisis periods, while remaining computationally tractable for the dense grid benchmarks. The ESS threshold and bootstrap block length sensitivities are evaluated over the entire sequence of 377 rolling windows.

9.1 Grid resolution and bandwidth sensitivity

Table 11 details the sensitivity of the adaptive exchange algorithm to the separation oracle’s grid density. The standard implementation uses a 21×21 grid (441 candidate states). Increasing the grid resolution from 11×11 to 81×81 increases the average

worst-case CVaR from approximately 4.46% to 4.53%, with limited objective variation beyond the 21×21 grid. Nevertheless, the average ℓ_1 allocation distance between the 21×21 and 81×81 solutions is approximately 0.063, indicating that objective stability does not imply complete allocation stability. Average solve time remains below one second for all examined grids. Table 7 reports one representative window wall-clock measurement covering kernel-matrix preparation within the solver routine, JuMP model construction, and HiGHS execution. Table 11 reports averages over 20 rolling-window instances. The two timing experiments therefore differ in both selected instances and aggregation and should not be compared as identical benchmarks.

Table 11 Grid size sensitivity: impact on solve time, active states, and objective

Grid resolution	Runtime (s)	Active states	Worst-case CVaR ₉₅ (%)	Dist. to base
11×11	0.19	3.20	4.46	0.1213
21×21	0.26	3.35	4.52	0.0627
41×41	0.51	3.75	4.53	0.0107
81×81	0.82	3.70	4.53	0.0

Table 12 reports sensitivity to the kernel bandwidth scalar $c \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$. As bandwidth increases, the kernel density estimate becomes smoother and less sensitive to local extreme observations, leading to a reduction in the worst-case CVaR (from 4.60% at $c = 0.5$ to 4.30% at $c = 2.0$). We retain $c = 1.0$ as the baseline rule-of-thumb bandwidth. The sensitivity results show that smaller bandwidths produce higher worst-case CVaR values, while larger bandwidths smooth local state effects.

Table 12 Bandwidth scalar sensitivity: impact on active states and objective

Scalar (c)	Active States	In-Sample Worst-Case CVaR ₉₅ (%)
0.5	3.65	4.60
0.75	3.35	4.56
1.0	3.35	4.52
1.5	3.45	4.38
2.0	3.30	4.30

9.2 Bandwidth orientation

Table 12 varies the scale of the bandwidth but not its orientation: the baseline rule is diagonal, $H_{\text{diag}} = T^{-1/3} \text{diag}(\hat{\sigma}_v^2, \hat{\sigma}_d^2)$, even though log-VIX and drawdown are correlated, with a mean within-window correlation of 0.70 over the 377 windows. A diagonal bandwidth orients the kernel along the coordinate axes and therefore averages over states that the joint distribution makes implausible. We repeat the full rolling experiment with the full-covariance rule $H_{\text{full}} = T^{-1/3} \hat{\Sigma}_y$, which carries the same scaling and adds only the off-diagonal term.

Table 13 shows that orientation does not matter here. The Sharpe ratio moves from 0.809 to 0.810, the mean worst-case conditional CVaR from 4.525% to 4.535%, the mean number of active states from 3.66 to 3.63 and turnover from 9.48% to 9.35%. The one quantity that responds is the effective sample size at the active

Table 13 Diagonal versus full-covariance bandwidth, on the same 377-window calendar as Table 4

Bandwidth	Sharpe	Turn. (%)	Wealth	Worst CVaR (%)	Active states	Mean ESS
H_{diag} (baseline)	0.809	9.48	25.04	4.525	3.66	83.0
H_{full}	0.810	9.35	25.40	4.535	3.63	64.3

states, which falls from 83.0 to 64.3: aligning the kernel with the principal axis of the state cloud concentrates the weights on fewer observations, as expected. That this sharpening leaves the allocation and the worst-case value essentially unchanged is itself informative, since it indicates that the binding states are determined by the geometry of the state domain rather than by the fine structure of the local weighting. The diagonal rule is retained for the baseline because it is the simpler of the two and nothing in this comparison favours the alternative.

9.3 ESS regularization and bootstrap block length

To address concerns about boundary scenario overfitting in regions of the state space with low empirical support, we evaluate an ESS-thresholded uncertainty set $\mathcal{U}_{\text{supp}} = \{\theta \in \mathcal{U} : \text{ESS}(\theta) \geq E_{\min}\}$. Table 14 shows that enforcing $E_{\min} = 10$ filters out over half of the candidate states (retaining 172.5 out of 441) without materially altering the worst-case CVaR (4.45%). However, requiring $E_{\min} \geq 40$ more aggressively restricts the admissible state space, which materially reduces the estimated worst-case conditional CVaR by excluding low-support boundary states (worst CVaR drops to 2.69%).

Table 14 Effective Sample Size (ESS) regularizer sensitivity

$E_{\min}$	Avg Retained States	In-Sample Worst-Case CVaR ₉₅ (%)
0.0	441.0	4.45
10.0	172.4	4.45
20.0	118.8	3.72
40.0	78.6	2.69
60.0	61.4	2.52
100.0	42.4	2.26

To understand whether this in-sample regularization translates to out-of-sample improvements, we run the complete rolling backtest pipeline across four selected ESS thresholds: $E_{\min} \in \{0, 10, 20, 40\}$. Table 15 reports the out-of-sample performance and active state statistics for each case. While $E_{\min} = 10$ successfully excludes over 60% of the grid (retaining an average of 39.3% of states) and excludes candidate grid states whose ESS falls below 10, the point estimates are economically similar, although no formal equality test is conducted. Aggressive regularization ($E_{\min} = 40$) slightly increases the out-of-sample Sharpe ratio but exposes the portfolio to a severe maximum drawdown (-45.94%). We retain $E_{\min} = 0$ as the baseline primary specification and report $E_{\min} \in \{10, 20, 40\}$ as support-regularized sensitivity analyses.

Finally, Table 16 verifies the robustness of our statistical inference to the chosen block length b in the paired circular block bootstrap. Across all block

Table 15 Out-of-sample performance across ESS thresholds ($E_{\min}$)

Thresh- old	Ret. (%)	Vol. (%)	Sharpe	Max DD (%)	Wealth	Turn. (%)	Mean ESS	Min. ESS	Grid ret. (%)
$E_{\min} = 0$	11.28	13.95	0.81	-41.69	25.04	9.48	83.0	2.2	100.0
$E_{\min} = 10$	11.04	13.92	0.79	-41.71	23.29	9.64	86.9	10.0	39.3
$E_{\min} = 20$	11.02	13.58	0.81	-40.84	23.46	11.27	92.5	20.0	27.5
$E_{\min} = 40$	10.98	13.26	0.83	-45.94	23.53	13.02	112.6	40.0	18.3

lengths from 6 to 24 holding periods, the standard error remains relatively stable, and the two-sided p -value for the Sharpe ratio difference against the Nominal CVaR benchmark never falls below the conventional 0.05 threshold, supporting the qualitative conclusion that the inference is not sensitive to the reported range of block lengths. The block-length sensitivity results use the same methodology as the primary bootstrap analysis. Remaining numerical differences across block lengths reflect both the altered dependence structure induced by the block length and Monte Carlo variability.

Table 16 Bootstrap block length sensitivity on inference

Block length b (21-day holding periods)	Bootstrap SE	p -value
6	0.0655	0.270
9	0.0658	0.275
12	0.0657	0.281
18	0.0618	0.243
24	0.0535	0.183

9.4 Sensitivity to the state-domain expansion

Because the robustness operator takes a worst case over $\mathcal{U}$, and $\mathcal{U}$ is the training-window box expanded by a margin δ beyond the observed extrema, that margin is not a minor implementation setting: it determines which states count as adverse. We therefore repeat the complete rolling experiment for $\delta \in \{0\%, 5\%, 10\%, 20\%\}$, where 10% reproduces the baseline. Estimation window, holding period, weight cap, return target, bandwidth rule and transaction costs are identical across the four runs, so the margin is the only quantity that varies. This separates robustness attributable to the continuous-state formulation from robustness induced by the particular choice of domain size.

Table 17 Sensitivity to the state-domain expansion margin δ . All four runs use the identical 377-window calendar; the last column is the fraction of windows in which at least one generated state lies on the boundary of the expanded box

Margin δ	Ret. (%)	Vol. (%)	Sharpe	Max DD (%)	Turn. (%)	Wealth	Mean ESS	Active states	Boundary (%)
0%	11.33	13.96	0.811	-41.77	9.51	25.40	83.3	3.75	48.8
5%	11.35	13.96	0.813	-41.71	9.51	25.63	84.5	3.62	48.5
10%	11.28	13.95	0.809	-41.69	9.48	25.04	83.0	3.66	49.3
20%	11.33	13.94	0.813	-41.45	9.20	25.44	82.9	3.62	49.1

Table 17 shows that the reported conclusions are insensitive to this choice. Across a fourfold change in the margin, and including the case $\delta = 0$ in which the

domain is exactly the observed training-window box, the Sharpe ratio varies by 0.005, terminal wealth by less than \$0.60, the average number of generated states by 0.13, and the average worst-case CVaR by 0.003 percentage points. The baseline $\delta = 10\%$ is therefore not what produces the results.

The last column is more interesting. Boundary activity is frequent but not universal: in roughly 49% of the rolling windows at least one generated stress state lies on the boundary of the expanded box, and that proportion is essentially unchanged as the boundary is moved outward. This suggests that conditional CVaR often increases towards the edge of the admissible state space, while interior worst-case states remain relevant in the remaining windows. Enlarging the margin moves the boundary without changing this structure, which explains why the performance statistics barely respond. It also sharpens the support question of Section 9: the states that drive the robust constraint are, by construction, those furthest from the data cloud, and therefore those estimated with the least local support.

9.5 Sensitivity to the expected-return target

The baseline sets the return constraint to the cross-sectional median of the estimated annualized asset means. Section 7 attributes the elevated volatility of Robust SIP to the interaction of return targeting, state conditioning and boundary effects, without separating them. This experiment separates them. We repeat the full rolling experiment with the target at the 25th, 50th and 75th percentiles of the estimated means, and once with the constraint set to the minimum attainable portfolio mean so that it is present but never binding. Everything else is held at the baseline.

Table 18 Sensitivity to the expected-return target. Q_p denotes the p th percentile of the estimated annualized asset means and Q_{50} is the baseline; the last column is the mean number of active states, on the same 377-window calendar as Table 4

Target	Ret. (%)	Vol. (%)	Sharpe	Max DD (%)	Turn. (%)	Wealth	N_{eff}	Active
Non-binding	10.84	14.00	0.774	-43.57	6.12	21.80	7.27	3.79
Q_{25}	10.91	14.02	0.778	-43.57	6.38	22.25	7.29	3.80
Q_{50}	11.28	13.95	0.809	-41.69	9.48	25.04	7.36	3.66
Q_{75}	13.07	14.54	0.899	-39.49	13.64	42.62	7.36	3.15

Table 18 reports the four specifications. Its Q_{50} row reproduces the baseline of Table 4 exactly, which serves as a check on the experiment. The first conclusion is negative and clear: the return target is not what makes Robust SIP volatile. Removing the constraint altogether leaves annualized volatility at 14.00% against 13.95% at the baseline, and across all four specifications volatility varies only between 13.95% and 14.54%. The elevated volatility relative to the CVaR benchmarks, which run near 12.2–12.4%, is a property of minimizing a worst-case conditional CVaR over a state domain, not a by-product of return targeting. Concentration behaves the same way, with N_{eff} confined to 7.27–7.36.

The second conclusion is more consequential. The target is a first-order driver of everything else. Raising it from Q_{50} to Q_{75} raises the Sharpe ratio from 0.809 to 0.899, terminal wealth from \$25.04 to \$42.62, and shallows the maximum drawdown

from -41.7% to -39.5% , at the cost of raising turnover from 9.48% to 13.64% . Within the Robust SIP specification, the Sharpe ratio is highly sensitive to the expected-return target, increasing from 0.809 at Q_{50} to 0.899 at Q_{75} . Because the benchmark strategies are not re-estimated under the alternative target levels, this sensitivity analysis should not be interpreted as establishing cross-strategy parity at Q_{75} . Instead, it shows that part of the observed performance profile depends materially on the return-target convention.

We do not adopt Q_{75} as the baseline, and we caution against reading this row as a recommendation. The four targets are compared after observing the full out-of-sample record, so selecting the best-performing one would be an in-sample choice presented as an out-of-sample result. What the experiment establishes is narrower and, for the purposes of this paper, more useful: a reader comparing Table 4 across strategies is also, implicitly, comparing a particular return-target convention, and the sensitivity of the outcome to that convention is larger than the differences between the strategies themselves. The interaction between the return constraint and the robust layer also runs in the expected direction: the mean number of active states falls from 3.79 to 3.15 as the target tightens, and the mean worst-case conditional CVaR rises from 4.46% to 4.94% , since a tighter return floor forces the allocation into states with less favourable tails.

9.6 Support restriction and state conditioning

Two further specifications isolate the roles of the support of the state domain and of the robust layer itself. Both are computed on the identical 377-window calendar and with the identical estimation window, weight cap, return target, bandwidth rule and transaction-cost convention as the baseline, so the only differences are the ones under study.

The first restricts the separation set to the empirical convex hull of the training states, so that the worst case is taken only over combinations of log-VIX and drawdown that the sample actually exhibits. Figure 16 shows the geometry for a representative window: the hull excludes the low-volatility/high-drawdown and high-volatility/low-drawdown corners entirely, and on average only 32.2% of the 441 candidate states survive. The second removes the robust layer and solves nominal CVaR under the kernel-weighted conditional distribution at the current state y_T , which is the prescriptive counterpart in the spirit of Bertsimas and Kallus [4].

Table 19 Support-restricted and state-conditioned specifications, on the same 377-window calendar as Table 4

Specification	Ret. (%)	Vol. (%)	Sharpe	Turn. (%)	Wealth
Robust SIP (baseline)	11.28	13.95	0.809	9.48	25.04
Robust SIP, hull-restricted	11.06	13.99	0.790	9.47	23.32
State-conditioned CVaR	10.05	13.01	0.773	35.74	17.82

Table 19 reports both. Neither variant improves on the baseline. Restricting the domain to the empirical hull leaves the maximum drawdown and the turnover essentially unchanged and lowers the Sharpe ratio slightly, from 0.809 to 0.790 . The low-support boundary states are therefore not the source of the risk-adjusted

shortfall documented in Section 7: removing them costs a little performance rather than recovering any. This is a useful negative result, since it separates the support question from the performance question.

The state-conditioned specification is the more informative comparison. It is dominated on every dimension reported here, with the lowest Sharpe ratio (0.773), the lowest terminal wealth (\$17.82) and an average turnover of 35.7%, nearly four times that of the baseline. Conditioning the distribution on the single currently observed state makes the allocation track short-lived movements in the state variables, which is precisely the instability that taking the worst case over a state domain suppresses. Figure 17 shows the resulting wealth paths. The robust layer therefore earns its place not by beating unconditional benchmarks on risk-adjusted terms, which it does not, but by stabilizing an allocation that pure conditioning renders excessively turnover-intensive.

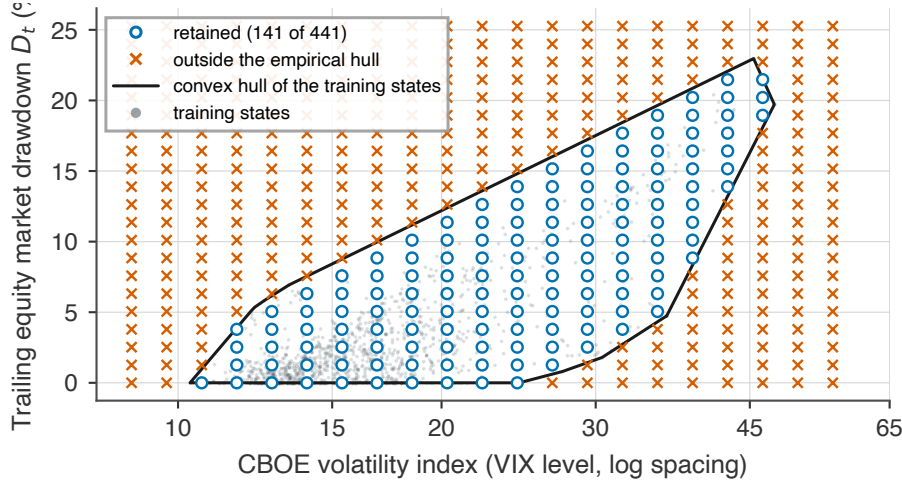


Fig. 16 Candidate states retained after restriction to the empirical convex hull of the training sample, for a representative rolling window. Circles mark retained states, crosses those excluded, and the solid line is the hull boundary

9.7 A regularized conditional comparator

The state-conditioned specification of Section 9.6 conditions on the single observed state y_T and nothing else, and it is fair to ask whether it is too fragile a comparator to be informative. We therefore embed it in a one-parameter family that shrinks the conditional law towards the unconditional one,

$$\hat{P}_t^{(\lambda)} = (1 - \lambda) \hat{P} + \lambda \hat{P}_{y_T}, \quad \lambda \in [0, 1], \quad (45)$$

and solves nominal CVaR under $\hat{P}_t^{(\lambda)}$. Here $\hat{P}$ places equal mass on the training sample and $\hat{P}_{y_T}$ carries the Nadaraya–Watson weights at the current state, so λ interpolates continuously between the two benchmarks already in the study. The rolling calendar and every estimation setting are those of the baseline.

Table 20 reports five values of λ . Two endpoints serve as an implementation check: $\lambda = 0$ reproduces the Nominal CVaR row of Table 4 exactly, and $\lambda = 1$

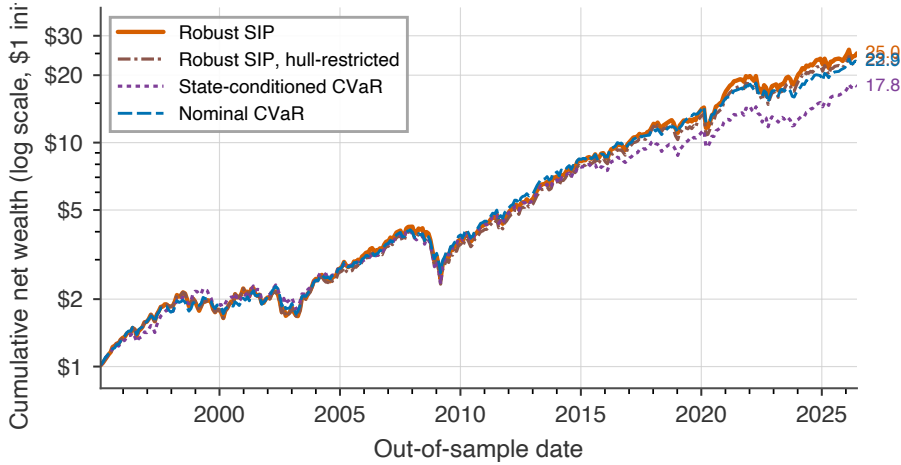


Fig. 17 Cumulative out-of-sample wealth of the support-restricted and state-conditioned specifications against the baseline and the strongest unconditional benchmark, net of 10 basis points per unit of turnover

Table 20 Nominal CVaR under the shrunk conditional law of Equation (45), on the same 377-window calendar as Table 4

λ	Ret. (%)	Vol. (%)	Sharpe	Max DD (%)	Turn. (%)	Wealth
0.00	10.76	12.23	0.880	-35.76	8.23	22.92
0.25	10.44	12.38	0.843	-38.73	16.34	20.57
0.50	10.23	12.53	0.816	-41.67	21.83	19.17
0.75	10.40	12.82	0.811	-41.76	28.05	20.03
1.00	10.05	13.01	0.773	-42.73	35.74	17.82

reproduces the state-conditioned row of Table 19 exactly. Between them, every reported quantity degrades monotonically in λ . The Sharpe ratio falls from 0.880 to 0.773, the maximum drawdown deepens from -35.8% to -42.7%, and average turnover rises from 8.2% to 35.7%, a factor of 4.3. No interior value of λ recovers what conditioning costs; regularizing the conditional law slows the deterioration without reversing it. The comparison in Section 9.6 is therefore not against a straw man, but against the best member of a family that contains it.

The diagnostic is not a small-sample artifact. The mean effective sample size at the conditioning state is 198.3 observations out of 1260, so the conditional distribution is estimated from a substantial fraction of the window. What destabilizes the allocation is not the precision of $\hat{P}_{y_T}$ at any one date but its movement between dates, and shrinking towards $\hat{P}$ damps that movement only in proportion to $1 - \lambda$.

Read against the baseline, the family also locates the contribution of the robust layer. Robust SIP attains a Sharpe ratio of 0.809, which sits between the $\lambda = 0.5$ and $\lambda = 0.75$ members, but it does so at an average turnover of 9.48%, close to the unconditional 8.2% and roughly a third of what those two members require, while accumulating more terminal wealth (\$25.04) than any member of the family. Taking a worst case over a state domain and shrinking towards the unconditional law are thus not two routes to the same place: the first buys state-awareness at almost unconditional trading cost, the second pays for it in turnover. This does not

overturn the central finding of Section 7, since the unconditional $\lambda = 0$ specification still records the highest Sharpe ratio in the table.

9.8 Sensitivity to the risk-free rate

Every Sharpe ratio reported so far uses a zero risk-free rate, which is difficult to defend over a sample running from 1994 to 2026. All five strategies are fully invested, so a constant rate is subtracted from every mean return alike; the ordering can nonetheless change, because a fixed subtraction penalizes a low-volatility strategy more than a high-volatility one. Table 21 recomputes the Sharpe ratios at 0, 2 and 4 percent per annum.

Table 21 Out-of-sample Sharpe ratios under a constant annual risk-free rate

Strategy	Ret. (%)	Vol. (%)	Sharpe $r_f = 0\%$	Sharpe $r_f = 2\%$	Sharpe $r_f = 4\%$
1/N	12.24	17.00	0.720	0.602	0.485
TC-MinVar	10.73	12.26	0.875	0.712	0.549
Nominal CVaR	10.76	12.23	0.880	0.717	0.553
Finite-Regime CVaR	11.15	12.38	0.901	0.739	0.577
Robust SIP	11.28	13.95	0.809	0.665	0.522

The ordering is identical at all three rates, so the conclusions of Section 7 do not depend on the zero-rate convention. The one effect worth noting runs mildly in favour of the proposed method: because Robust SIP carries the highest volatility among the optimized strategies, its Sharpe ratio declines most slowly as the rate rises, and its shortfall against the best benchmark narrows from 0.092 at $r_f = 0\%$ to 0.055 at $r_f = 4\%$. This remains a shortfall, and Section 8 has already shown that differences of this size are not statistically distinguishable from zero. We report a constant rate rather than a realized Treasury-bill series, so this is a check that the ranking survives a non-zero rate, not a measurement of realized excess returns.

9.9 Allocation instability and turnover constraints

A notable property of the continuous-state robust formulation is the near-flatness of the worst-case objective surface around the optimum. Because multiple asset combinations can achieve nearly identical worst-case CVaR, the optimal allocation w^* may exhibit instability across adjacent rolling windows. The grid-refinement analysis of Table 11 provides indirect evidence of this flatness: refining the separation grid from 21×21 to 81×81 changes the worst-case CVaR by less than 0.02 percentage points while displacing the optimal allocation by $\|w_{21} - w_{81}\|_1 \approx 0.063$. Objective stability therefore coexists with non-trivial allocation instability, which is consistent with the elevated mean holding-period turnover of 9.48% (median 5.0%) reported in Table 4 (11.37 bps approximate annual transaction-cost drag). While the focus of this paper is unconstrained risk minimization, practical implementations could penalize this instability, for instance by incorporating an ℓ_1 proximity regularizer $\lambda\|w - w_{\text{prev}}\|_1$ into the master LP.

10 Discussion and conclusion

The proposed state-robust CVaR framework provides a principled and interpretable method for incorporating continuous market covariates into portfolio optimization without imposing artificial discrete regime boundaries. In the reported approximately 31.5-year empirical experiment on US industry portfolios, the continuous-state strategy achieved greater terminal wealth (\$25.04) than TC-MinVar (\$22.65) and nominal CVaR (\$22.92), but lower wealth than naive $1/N$ (\$29.25) and finite-regime CVaR (\$25.68). It also exhibited lower Sharpe ratios and deeper maximum drawdowns than the nominal and finite-regime CVaR benchmarks, and paired circular block percentile bootstrap testing does not provide sufficient evidence to distinguish these Sharpe-ratio differences from zero at conventional significance levels. Thus, the empirical findings do not establish superior risk-adjusted or crisis performance. Two specification experiments qualify how this shortfall should be read. It is not produced by the return-target convention in the way one might assume: removing the return constraint entirely leaves annualized volatility unchanged at 14.00%, so the elevated volatility is a property of worst-case conditional CVaR minimization rather than of return targeting. The Sharpe ratio, by contrast, is highly sensitive to that convention, rising to 0.899 under a 75th-percentile target. Since the benchmarks are not re-estimated at that target, this does not establish parity with them; it establishes that the return-target convention accounts for a material part of the performance profile, and that a comparison across strategies at one fixed convention is correspondingly less decisive than it appears.

The results do not establish that continuous-state robustification dominates nominal or finite-regime CVaR in risk-adjusted performance. The evidence supports a more specific conclusion. First, continuous state dependence can be incorporated into a convex robust portfolio model without requiring discrete regime boundaries. Second, despite the continuum of state-indexed constraints, the finite grid approximation is computationally sparse, and sparse in a way that does not degrade with resolution: an average of 3.66 out of 441 candidate blocks is generated at the baseline, and in a matched comparison against the extensive-form dense LP over 40 windows the average stays between 3.27 and 3.42 as the candidate set grows eightfold from 121 to 961 states, while dense solution time grows by a factor of 17.1 and exchange time by a factor of 3.5. It is this stability, rather than the speedup on any one instance, that makes a fine discretization of a continuous state space affordable. Third, robustness across the state domain substantially stabilizes the highly reactive allocations produced by pure state-conditioned CVaR, which turns over 35.7% of the portfolio per rebalancing against 9.48% for the robust formulation, and no intermediate degree of shrinkage towards the unconditional law recovers the difference. The principal contribution is therefore methodological and computational, with state robustness acting as a regularizing mechanism rather than as a guarantee of superior investment performance.

Several methodological considerations and future extensions warrant discussion. First, while the two-dimensional state space (log-VIX and drawdown) remains computationally tractable, kernel density estimation and Cartesian grid search deteriorate rapidly as state dimension increases ($M \geq 3$). Future research should explore dimension-reduction techniques, such as autoencoder latent spaces or conditional generative models, to scale continuous-state robustification. Second, in two dimensions, global grid evaluation provides an efficient separation oracle, whereas

in higher dimensions, certified global branch-and-bound or interval arithmetic could provide certified continuous optimality bounds. Third, our empirical tracking of Effective Sample Size revealed that the minimum, across rolling windows, of the within-window mean ESS over active states is 2.18, indicating that some windows contain collectively low-support active stress states. Incorporating explicit ESS-thresholded uncertainty domains $\mathcal{U}_{\text{supp}} = \{\theta \in \mathcal{U} : \text{ESS}(\theta) \geq E_{\min}\}$ represents a natural regularizer to prevent boundary scenario overfitting. Fourth, integrating explicit turnover regularization directly into the master LP would further suppress transaction costs in higher-friction market environments. The framework combines non-parametric conditional risk estimation with a continuous-state semi-infinite formulation, while the empirical implementation solves a finite grid-restricted approximation efficiently through adaptive constraint generation.

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Data Availability: The Kenneth French 30 Industry Portfolio returns [16] and CBOE VIX observations [9] used in this study were obtained from their publicly accessible source pages. Source URLs, retrieval information, preprocessing code, and the aligned research dataset are documented in the accompanying repository, subject to the original providers' terms of use. Reference [8] cites the Zenodo concept DOI, which always resolves to the most recent archived version; the version-specific DOI and the Git commit identifier of the snapshot corresponding to the results reported here are recorded in the release notes of that version. The GitHub repository may contain later development versions [8].

Author Contributions: Madani Bezoui: Conceptualization, Methodology, Software, Writing - original draft. Thiziri Sifaoui: Formal analysis, Validation, Writing - review and editing. Ahcene Bounceur: Supervision, Writing - review and editing.

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